PD-Prune: Disaggregated Prefill-Decode Pruning for Efficient Agentic LLM Serving

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Abstract

Agentic LLM serving often expands one committed context into multiple candidate continuations, but only one is retained. Under prefill–decode disaggregation, every admitted candidate may incur residual prefill and KV transfer before existing decode-time pruning can act. Prefix caching removes repeated work on the shared trunk, but does not prevent wasted work on residuals that are later discarded. We present PD-Prune, a cross-stage pruning framework that moves branch filtering to the earliest stage with sufficient information. Before prefill, PD-Prune combines dual-side KV residency with a lightweight score to reject low-value residuals or admit them directly. Uncertain residuals are refined by a short decode-side probe, while a copy-on-write context tree shares the trunk, reclaims aborted pages, and retains the committed spine. We evaluate PD-Prune on Qwen3 models and reasoning workloads against vLLM disaggregated serving. On GSM8K, prefill-side admission reduces computed prefill by 23% at matched committed-answer accuracy. On MATH-500, a 128-token probe reduces wall-clock to $0.69\times$ while maintaining reasoning quality. Spine retention reduces median peak KV usage to $0.72\times$.

1 Introduction

Agentic LLM applications interleave generation with branching. A coding agent proposes several edits or tool calls for one repository context and executes one [36, 34, 30]; the same pattern appears when an inference-time controller expands a Tree-of-Thoughts frontier or a self-consistency pool [35, 31, 32]. At each step a committed trunk fans out into k residuals, and one becomes the next spine while the rest are discarded [8, 12, 17].

Branching improves search coverage, but it also multiplies serving cost. Under prefill–decode (PD) disaggregation, prompt processing and token generation run on separate instances, and the KV pages produced by prefill must cross a connector before decode can begin [42, 25]. Every admitted residual may therefore pay residual prefill, KV transfer, and decode before it is discarded, and that cost accumulates as the agent repeats the fan-out.

Prior work improves efficiency mainly along two axes, and neither removes this cost. Decode-time path pruners act after generation has started [17, 26, 10, 4], so residual prefill and the connector transfer have often already been paid. Prefix caching shares the trunk when its pages are resident [14, 41], but neither rejects a residual nor controls which pages cross the connector. Even a fully shared trunk still pays for each residual suffix that is later discarded. Pruning therefore remains *stage-local*: each stage can discard only work that has already reached it. The goal is a disaggregated design across prefill and decode, a hardware-algorithm co-design for efficient agentic LLM serving.

Achieving this goal, however, presents two key challenges.

Challenge 1: Disaggregated residency. Prefill and decode maintain KV on separate instances, making cache residency inherently stage-specific. A page resident on the prefill instance may still be absent from decode and require connector transfer, while decode-side residency can eliminate that transfer. The challenge is therefore to efficiently coordinate residency information across the disaggregated prefill and decode instances.

Challenge 2: Prefill-side admission. Once the prefill kernel is launched, the uncached residual suffix is computed and decoder-missing KV pages enter the transfer path. Admission therefore needs to be decided beforehand, using only the residual text and the two residency lengths. The challenge is to enable reliable prefill-side admission under such limited information with minimal serving-system overhead.

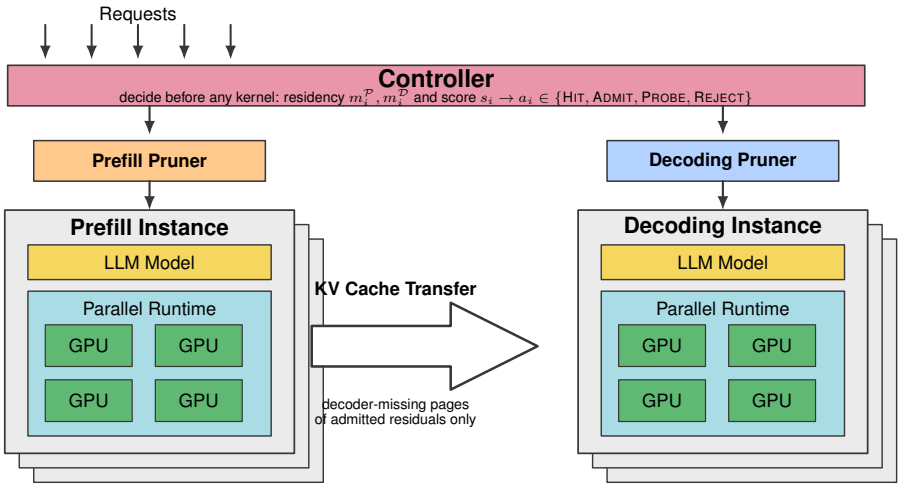


Figure 1: **PD-Prune runtime.** The controller reads the residency of each residual on the prefill instance ($m_i^{\mathcal{P}}$) and the decode instance ($m_i^{\mathcal{D}}$) together with a lightweight model-free heuristic score s_i (the prefill-side score), and assigns an action $a_i \in \{\text{HIT}, \text{ADMIT}, \text{PROBE}, \text{REJECT}\}$ before any kernel runs. The prefill instance computes only the non-resident suffix of admitted and probed residuals; the connector carries only the pages the decode instance lacks. The decode-side pruner extends a probed residual beyond t_0 tokens only if its generated prefix clears τ_{dec}. After commit, only the spine $x_\star$ is retained.

Disaggregated pruning. Our key insight is to disaggregate pruning across the prefill and decode stages (Figure 1). PD-Prune admits or rejects residuals before prefill from dual-side KV residency and a lightweight score (§4.1), refines uncertain residuals with a short decode-side probe (§4.2), and uses a pruning-aware copy-on-write context tree to share the trunk, reclaim aborted pages, and retain the committed spine (§4.4). Existing decode-time pruners operate unchanged on the admitted set.

Results. On Qwen3 models, prefill-side admission reduces computed prefill by 23% at matched committed-answer accuracy; on MATH-500, a 128-token probe restores coverage to $20/32$ at $0.69\times$ the wall-clock of the full fan-out; and spine retention reduces median peak KV usage to $0.72\times$ that of APC (§5).

Contributions. Our contributions are summarized as follows:

- We formulate pruning in disaggregated agentic serving as a per-residual, cross-stage decision problem, and characterize cache regimes by trunk-amplification multiplicity on the prefill instance and on the connector (§3).
- We propose *disaggregated pruning* and instantiate it in PD-Prune with prefill-side admission and decode-side probing; lightweight KV state management supports branch reclamation and spine retention (§4).
- We demonstrate that prefill-side admission composes with three existing decode-time pruning methods and reduces computed prefill, and that decode-side probing mitigates the coverage loss of aggressive prefill-side rejection, while preserving committed-answer accuracy (§5).

2 Related Work

Inference-time search and path pruning. Tree-of-Thoughts, self-consistency, and Monte Carlo tree search expand many partial solutions and then select [35, 31, 39, 32]. Decode-time path pruners cut that cost after generation has started: ESC stops once a window agrees [17], Speculative Rejection halts low-reward prefixes at a horizon [26], DPTS early-stops children after a mini-step [10], and STOP trains an internal futility token [4]. Equiv-Pruner merges semantically equivalent steps so that textually distinct but mathematically identical actions are not explored twice [21]. All of these decide from generated or written text and are agnostic to where the KV cache was computed. PD-Prune is complementary: it adds a pre-kernel decision and a decode-side probe, while these decode-time pruners operate unchanged on the admitted set.

Prefix sharing and PD disaggregation. Orca schedules at iteration granularity [37]; vLLM pages the KV cache and hashes full blocks for APC [14];

SGLang shares prefixes through a radix tree [41]; Sarathi chunks prefills to co-schedule them with decode [2, 3]. DistServe and Splitwise place prefill and decode on separate instances and move KV across a connector [42, 25]. These mechanisms determine whether a trunk is shared and how pages move, but they serve every request they receive. Whether k siblings share the trunk depends on the cache regime (§3.2); even when the trunk is fully reused, each residual's uncached suffix is still computed and transferred. PD-Prune uses residency on both instances as an admission input and treats the connector as the channel whose volume admission reduces.

Agentic serving, retention, and copy-on-write. Autellix and Hermes schedule at program granularity [23, 22]; ThunderAgent and SAGA optimize workflows [13, 11]; Parrot, Ayo, and Pythia exploit application structure [20, 27, 38]; JITServe allocates bandwidth under heterogeneous SLOs [40]. InferCept, Continuum, and MORI retain KV across tool calls [1, 16, 33]; ConServe disaggregates prefill and decode scheduling at conversation granularity from observed agent behavior [9]; Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]. ForkKV is closest in mechanism: it shares a base KV cache copy-on-write across LoRA agents [29]. CachePrune edits attention state inside one request [28]; speculative decoding drafts *output* tokens rather than a future prompt prefix [15, 6, 18]. These systems place, schedule, or retain contexts that will be served; ConServe in particular disaggregates scheduling but not pruning. PD-Prune targets the complementary decision of which sibling continuations should be served at all, and at which PD stage that decision is made. Its copy-on-write context tree is the state-management mechanism that supports those decisions for siblings of one agent step.

Appendix E expands the literature survey.

3 Problem Formulation

We model one agent step as a branching step served by a prefill instance $\mathcal{P}$ and a decode instance $\mathcal{D}$ connected by a KV connector. Table 7 in the appendix summarizes the notation.

Definition 1 (Branching step)**.** A parent u holds a committed trunk x_u with $|x_u|{=}L$. A fan-out is a set of candidate continuations. We call the candidate-specific suffix ρ_i a *residual*, with $|\rho_i|{=}\ell_i$; candidate i is the sequence $x_u\|\rho_i$. Index 0 denotes the highest-ranked candidate produced by the branching controller (e.g., the greedy continuation). Commit selects one continuation $\star$ and leaves the spine $x_\star = x_u\|\rho_\star$, which becomes the trunk of the next step.

With b bytes of KV per token, three occupancies characterize one step:

$$\begin{aligned} M_{\text{clone}} &= b\big(kL + \textstyle\sum_i \ell_i\big), \\ M_{\text{CoW}} &= b\big(L + \textstyle\sum_i \ell_i\big), \\ M_{\text{spine}} &= b\big(L + \ell_\star\big). \end{aligned} \tag{1}$$

M_{clone} arises only when every request materializes its own trunk; M_{CoW} holds one shared trunk and all residuals; M_{spine} is the occupancy after uncommitted residuals are freed.

3.1 Residency on Two Instances

Paged engines index full blocks of P tokens by a chained key $h_j = \text{hash}(h_{j-1}, \text{block}_j, \text{extra})$ [14, 41], so a block is matched only together with its entire parent prefix. For an instance $\mathcal{X} \in \{\mathcal{P}, \mathcal{D}\}$ we define the resident prefix of the trunk and of residual i as

$$\begin{aligned} c_u^{\mathcal{X}} &= \max\{jP \le L :\ x_u[0{:}jP) \sqsubseteq \mathcal{X}\}, \\ m_i^{\mathcal{X}} &= \max\{jP \le \ell_i :\ x_u\|\rho_i[0{:}jP) \sqsubseteq \mathcal{X}\}, \end{aligned} \tag{2}$$

where $z \sqsubseteq \mathcal{X}$ means that every page of z is physically resident on $\mathcal{X}$, and $m_i^{\mathcal{X}}{=}0$ whenever $c_u^{\mathcal{X}}{<}L$, because chained keys make a residual page unreachable without its full trunk. For exposition we take L to be a multiple of P; otherwise the partial trunk page is charged to each residual. Three distinctions matter. (i) *Parent-aware match:* $m_i^{\mathcal{X}}$ is defined on $x_u\|\rho_i$, not on ρ_i alone. (ii) *Index presence versus residency:* an index entry whose block has been evicted does not count; $m_i^{\mathcal{X}}$ counts physically resident pages only. (iii) *Instance locality:* pages resident on $\mathcal{P}$ are not visible on $\mathcal{D}$, so $m_i^{\mathcal{P}}$ and $m_i^{\mathcal{D}}$ generally differ.

3.2 Cache Regimes

Prefix caching alone does not determine whether a fan-out pays for the trunk once or k times. Let

$$\bar{L}_{\mathcal{X}} = L - c_u^{\mathcal{X}}, \qquad \mathcal{X} \in \{\mathcal{P}, \mathcal{D}\}, \tag{3}$$

be the trunk tokens missing on instance $\mathcal{X}$. We define $\nu_u^{\mathcal{P}}$ as the number of distinct materializations of $\bar{L}_{\mathcal{P}}$ on the prefill instance and $\nu_u^{\mathcal{D}}$ as the number of distinct connector transfers of $\bar{L}_{\mathcal{D}}$. Thus

$$W^{\text{trunk}} = \nu_u^{\mathcal{P}} \bar{L}_{\mathcal{P}}, \qquad X^{\text{trunk}} = \nu_u^{\mathcal{D}} \bar{L}_{\mathcal{D}}, \tag{4}$$

with $\nu_u^{\mathcal{X}}=0$ if $\bar{L}_{\mathcal{X}}=0$, and $1 \le \nu_u^{\mathcal{X}} \le k$ otherwise. We additionally let $\sigma_u \in \{0,1\}$ denote a publication barrier: $\sigma_u=1$ delays sibling admission until one reusable trunk copy is published. A cache regime is therefore characterized by

$$\mathcal{R}_u = (\nu_u^{\mathcal{P}}, \nu_u^{\mathcal{D}}, \sigma_u). \tag{5}$$

The prefill-side regimes are special cases:

$$\begin{aligned} R_1 \text{ (cold, concurrent): } & c_u^{\mathcal{P}}=0,\ \sigma_u=0, \\ & 1 \le \nu_u^{\mathcal{P}} \le k, \\ R_2 \text{ (cold, serialized): } & c_u^{\mathcal{P}}=0,\ \sigma_u=1,\ \nu_u^{\mathcal{P}}=1, \\ R_3 \text{ (warm): } & c_u^{\mathcal{P}}=L,\ W^{\text{trunk}}=0, \\ R_4 \text{ (explicit fork): } & \sigma_u=0,\ \nu_u^{\mathcal{P}}=1. \end{aligned} \tag{6}$$

Hence R_1 gives $L \le W^{\text{trunk}} \le kL$, R_2 gives $W^{\text{trunk}}=L$ but introduces a serialization barrier, and R_4 gives $W^{\text{trunk}}=L-c_u^{\mathcal{P}}$ without serialization. Connector behavior is characterized independently by $\nu_u^{\mathcal{D}}$: request-local transfer may have $\nu_u^{\mathcal{D}}=k$, whereas shared-trunk aliasing gives $\nu_u^{\mathcal{D}}=1$.

For $\bar{L}_{\mathcal{X}} > 0$, $\nu_u^{\mathcal{X}}$ is the trunk-amplification factor. Relative to independent fan-out, sharing saves $(k-\nu_u^{\mathcal{P}})\bar{L}_{\mathcal{P}}$ prefill tokens and $(k-\nu_u^{\mathcal{D}})\bar{L}_{\mathcal{D}}$ connector tokens. All GPU admission experiments in §5 use R_3 on $\mathcal{P}$, so their prefill savings are attributable to residual admission rather than trunk sharing. After commit, the spine is the logical trunk of the next step; its KV residency on $\mathcal{P}$ and $\mathcal{D}$ is restored according to the underlying KV synchronization policy.

3.3 Objective

Each residual receives an action a_i from a set $\mathcal{A}$, inducing computed prefill W_i, connector tokens X_i, and decode tokens d_i; we write $W=\sum_i W_i$, $X=\sum_i X_i$, and $D^+=\sum_i d_i$. We formulate the design objective as

$$\begin{aligned} \min_{\pi}\ & \mathbb{E}\big[T_{\text{pre}}(W) + \mu X + T_{\text{dec}}(D^+)\big] \\ \text{s.t. }\ & \mathbb{P}[\text{committed answer correct}] \ge Q_{\min}, \\ & M_{\text{peak}} \le M_{\text{pool}}, \qquad a_0 \ne \textsc{Reject}, \end{aligned} \tag{7}$$

where μ is the per-token transfer time. The defining constraint is informational: a decision taken before prefill may depend only on $(c_u^{\mathcal{P}}, c_u^{\mathcal{D}}, m_i^{\mathcal{P}}, m_i^{\mathcal{D}})$ and the residual text, whereas a decision taken on $\mathcal{D}$ may additionally read generated tokens. Decode-time pruners act only at the second point; PD-Prune adds the first and coordinates the two. PD-Prune instantiates this objective with a lightweight threshold-based policy.

4 PD-Prune: Cross-Stage Pruning

Algorithm 1 instantiates the cross-stage policy. §4.1–§4.4 describe admission, probing, accounting, and KV state; Appendix B.6 gives the multi-step pipeline.

4.1 Pre-Kernel Admission

For each residual the controller observes $o_i = (m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, s_i)$, where $m_i^{\mathcal{X}}$ is the parent-aware resident prefix of (2) and $s_i = \text{Score}_{\text{pre}}(\rho_i) \in [0,1]$ is a lightweight model-free heuristic score computed from the residual text in $O(\ell_i)$ CPU time. We refer to s_i as the prefill-side score. Our scorer combines repetition detection and syntactic checks (Appendix B.4); a learned scorer can replace it without changing the policy.

Definition 2 (Admission action)**.** Given thresholds $\tau_{\text{pre}} \le \tau_{\text{dec}}$, residual i receives the first matching action in

$$a_i = \begin{cases} \textsc{Hit} & m_i^{\mathcal{D}} = \ell_i, \\ \textsc{Admit} & i=0 \text{ or } s_i \ge \tau_{\text{dec}}, \\ \textsc{Reject} & s_i < \tau_{\text{pre}}, \\ \textsc{Probe} & \text{otherwise.} \end{cases} \tag{8}$$

HITrequires residency on the *decode* instance, since a page resident only on $\mathcal{P}$ must still be transferred. Because $i=0$ is caught by the first two cases, $a_0 \in \{\textsc{Hit}, \textsc{Admit}\}$: the top-ranked residual can be neither rejected nor probed away. Residuals in the uncertainty band $[\tau_{\text{pre}}, \tau_{\text{dec}})$ are probed. Setting $\tau_{\text{pre}}=\tau_{\text{dec}}$ empties the band and yields aggressive prefill-side rejection ("Blanket"); replacing PROBEby ADMITyields *reject-only admission*. Let $e_i=[\text{Score}_{\text{dec}}(y_i) \ge \tau_{\text{dec}}]$ indicate that the probe y_i of a probed residual is extended (§4.2). The action determines the per-residual costs

$$W_i = \begin{cases} 0 & \textsc{Hit}, \textsc{Reject}, \\ \ell_i - m_i^{\mathcal{P}} & \textsc{Admit}, \\ \ell_i - m_i^{\mathcal{P}} + e_i W_i^{\text{ext}} & \textsc{Probe}, \end{cases} \tag{9}$$

$$X_i = (\ell_i - m_i^{\mathcal{D}})\,[a_i \in \{\textsc{Admit}, \textsc{Probe}\}], \tag{10}$$

$$d_i = \begin{cases} 0 & \textsc{Reject}, \\ D & \textsc{Hit}, \textsc{Admit}, \\ t_0 + e_i(D - t_0) & \textsc{Probe}, \end{cases} \tag{11}$$

where D is the decode budget, t_0 the probe length, and W_i^{ext} the tokens recomputed when a probe is extended (§4.2). The trunk is aliased (§4.4)

Table 1: Scores of the heuristic scorers on the three residual classes of the evaluation mix. $\mathrm{Score}_{\mathrm{pre}}$ reads the residual; $\mathrm{Score}_{\mathrm{dec}}$ reads a probe that has not yet produced (malformed) or has produced (valid) a finished answer marker.

	repetitive	malformed	valid
$\mathrm{Score}_{\mathrm{pre}}(\rho_i)$	0.02	0.22	1.00
$\mathrm{Score}_{\mathrm{dec}}(y_i)$	0.04	0.16	0.92

in the explicit-fork regime $\nu_u^{\mathcal{P}}{=}\nu_u^{\mathcal{D}}{=}1$, $\sigma_u{=}0$ (§3.2), so one fan-out contributes $W^{\mathrm{trunk}}{=}\bar{L}_{\mathcal{P}}$ and $X^{\mathrm{trunk}}{=}\bar{L}_{\mathcal{D}}$ rather than one copy per sibling. A decode-time pruner (ESC, Speculative Rejection, DPTS) may further shorten d_i for any admitted or extended residual.

4.2 Decode-Side Probe

A probed residual is prefilled, transferred, and decoded for t_0 tokens; the decode-side pruner then scores the generated prefix, $s_i' = \mathrm{Score}_{\mathrm{dec}}(y_i)$, and extends the residual to D only if $s_i' \geq \tau_{\mathrm{dec}}$. The two thresholds read different text: τ_{pre} reads the residual before generation, and τ_{dec} reads model output. They are therefore calibrated separately. A useful pair satisfies the design condition

$$\sup_{\rho\in\mathcal{C}_{\mathrm{rej}}} s(\rho) < \tau_{\mathrm{pre}} \leq \inf_{\rho\notin\mathcal{C}_{\mathrm{rej}}} s(\rho), \qquad (12)$$
$$\sup_{y\in\mathcal{C}_{\mathrm{stop}}} s'(y) < \tau_{\mathrm{dec}},$$

where $\mathcal{C}_{\mathrm{rej}}$ is the class of residuals that should never be prefilled (here, repetitive residuals) and $\mathcal{C}_{\mathrm{stop}}$ the class of probes that should not be extended. Table 1 lists the scores of our heuristic scorers on the three residual classes of our workload; with $(\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}}) = (0.15, 0.45)$ both conditions hold. Because the heuristics assign near-constant scores per class, this separation is specific to our workload; a learned scorer can replace them on natural agent traces (§6).

Extension cost. If the probe's pages remain resident, extension continues decoding without recomputation and $W_i^{\mathrm{ext}}{=}0$. Our prototype issues the extension as a continuation request; when the probe's pages have been evicted in the meantime, the engine recomputes them, so $0 \leq W_i^{\mathrm{ext}} \leq \ell_i + t_0$. This term is negligible at $k \leq 8$ but substantial at $k{=}16$ in our runs (Appendix D.5), and pinning probe pages until the extension decision is an engineering fix we leave to future work.

Algorithm 1 PD-Prune (one fan-out). Actions follow (8); costs follow (9)–(11).

Require: trunk x_u; residuals $\{\rho_i\}$; residency on $\mathcal{P}/\mathcal{D}$; $(\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}}, t_0, D)$
Ensure: spine $x_\star$; (W, X, D^+)

Phase I: Admit ▷ decide before any kernel
$a_i \leftarrow \pi(m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, \mathrm{Score}_{\mathrm{pre}}(\rho_i))$ for all i
$\mathcal{A} \leftarrow \{i : a_i \in \{\textsc{Admit}, \textsc{Probe}\}\}$; $\mathcal{U} \leftarrow \{i : a_i{=}\textsc{Probe}\}$
$\mathrm{Fork}(x_u, \mathcal{A})$ ▷ alias trunk; residual pages only

Phase II: Prefill & Ship ▷ pay only what $\mathcal{P}/\mathcal{D}$ miss
$\mathrm{Prefill}_{\mathcal{P}}(x_u, \{\rho_i\}_{i\in\mathcal{A}})$ ▷ $\mathcal{P}$-miss; trunk once
$\mathrm{Ship}(x_u, \{\rho_i\}_{i\in\mathcal{A}})$ ▷ $\mathcal{D}$-miss

Phase III: Probe & Commit ▷ refine on $\mathcal{D}$, keep the spine
for all $i \in \mathcal{U}$ **do**
  $y_i \leftarrow \mathrm{Dec}(x_u \| \rho_i;\, t_0)$
  if $\mathrm{Score}_{\mathrm{dec}}(y_i) < \tau_{\mathrm{dec}}$ **then**
    $\mathrm{Abort}(\rho_i)$; $\mathcal{A} \leftarrow \mathcal{A} \setminus \{i\}$
  end if
end for
$\mathcal{S} \leftarrow \{i : a_i{=}\textsc{Hit}\} \cup \mathcal{A}$
$D^+ \leftarrow \mathrm{Prune}(\mathcal{S}; D)$; $\mathrm{Commit}(x_\star)$; $\mathrm{Abort}(\mathcal{S} \setminus \{\star\})$
return $x_\star, W, X, D^+$

4.3 Work and Transfer Accounting

We compare PD-Prune with a baseline that serves the same fan-out without admission.

Assumption 1 (Baseline)**.** The baseline admits every residual under a cache regime $(\nu_{\mathrm{B}}^{\mathcal{P}}, \nu_{\mathrm{B}}^{\mathcal{D}}, \sigma_{\mathrm{B}})$ (§3.2). It computes $\nu_{\mathrm{B}}^{\mathcal{P}}\bar{L}_{\mathcal{P}}$ trunk tokens and $\ell_i{-}m_i^{\mathcal{P}}$ tokens for each residual i, and transfers at least the pages each request lacks on $\mathcal{D}$,

$$X_{\mathrm{B}} \geq \nu_{\mathrm{B}}^{\mathcal{D}}\bar{L}_{\mathcal{D}} + \textstyle\sum_i(\ell_i - m_i^{\mathcal{D}}). \qquad (13)$$

Proposition 1 (Accounting)**.** *Fix a fan-out and a cache state, and let* $\mathcal{R}{=}\{i : a_i{=}\textsc{Reject}\}$, $\mathcal{H}{=}\{i : a_i{=}\textsc{Hit}\}$, *and* $\mathcal{E}{=}\{i : a_i{=}\textsc{Probe}, e_i{=}1\}$. *Let* $(\nu_u^{\mathcal{P}}, \nu_u^{\mathcal{D}})$ *be the multiplicities* PD-Prune *uses on that fan-out. Under Assumption 1,*

$$W_{\mathrm{B}} - W = \Delta_{\mathrm{trunk}} + \Delta_{\mathrm{adm}} - \Delta_{\mathrm{ext}}, \qquad (14)$$
$$X_{\mathrm{B}} - X \geq (\nu_{\mathrm{B}}^{\mathcal{D}} - \nu_u^{\mathcal{D}})\bar{L}_{\mathcal{D}} + \textstyle\sum_{i\in\mathcal{R}}(\ell_i - m_i^{\mathcal{D}}), \qquad (15)$$

where Δ_{trunk} *is trunk amplification,* Δ_{adm} *is residual admission, and* Δ_{ext} *is probe overhead:*

$$\Delta_{\mathrm{trunk}} = (\nu_{\mathrm{B}}^{\mathcal{P}} - \nu_u^{\mathcal{P}})\bar{L}_{\mathcal{P}},$$
$$\Delta_{\mathrm{adm}} = \textstyle\sum_{i\in\mathcal{R}\cup\mathcal{H}}(\ell_i - m_i^{\mathcal{P}}), \qquad (16)$$
$$\Delta_{\mathrm{ext}} = \textstyle\sum_{i\in\mathcal{E}} W_i^{\mathrm{ext}}.$$

Proof. Sum (9) over i and add $W^{\mathrm{trunk}}{=}\nu_u^{\mathcal{P}}\bar{L}_{\mathcal{P}}$. The baseline charges $\nu_{\mathrm{B}}^{\mathcal{P}}\bar{L}_{\mathcal{P}}$ and $\ell_i - m_i^{\mathcal{P}}$ for every i, while PD-Prune charges $\nu_u^{\mathcal{P}}\bar{L}_{\mathcal{P}}$, charges

the residual term only for $i \in \mathcal{A}$, and adds W_i^{ext} for $i \in \mathcal{E}$. The residuals outside $\mathcal{A}$ are exactly $\mathcal{R} \cup \mathcal{H}$, which gives Δ_{trunk} and Δ_{adm}. For transfer, add $X^{\text{trunk}} = \nu_u^{\mathcal{D}} \bar{L}_{\mathcal{D}}$ to (10). HITresiduals satisfy $\ell_i - m_i^{\mathcal{D}} = 0$, so they contribute on neither side. The baseline lower bound in (13) therefore exceeds PD-Prune's transfer by at least the trunk-transfer gap $(\nu_{\text{B}}^{\mathcal{D}} - \nu_u^{\mathcal{D}})\bar{L}_{\mathcal{D}}$ and the rejected residuals' decoder-missing pages. □

The decomposition separates three contributions: trunk-amplification reduction, residual admission, and probe overhead. Δ_{trunk} is zero when the baseline and PD-Prune materialize the missing trunk equally often, including every warm-trunk comparison (R_3, where $\bar{L}_{\mathcal{P}} = 0$), and it is positive when concurrent siblings recompute the trunk (R_1, $\nu_{\text{B}}^{\mathcal{P}} > \nu_u^{\mathcal{P}}$). Δ_{adm} does not depend on the regime, and Δ_{ext} is a cost. On the connector, (15) likewise separates trunk-transfer deduplication, $(\nu_{\text{B}}^{\mathcal{D}} - \nu_u^{\mathcal{D}})\bar{L}_{\mathcal{D}}$, from the admission saving on rejected residuals. Decode-time pruners change only D^+: since they act on generated tokens of residuals that have already been prefilled, W and X do not depend on the pruner. We decompose the stage-local work of a fan-out as

$$C_{\text{fan}} = C_{\text{pre}}(W) + C_{\text{transfer}}(X) + C_{\text{dec}}(D^+) + C_{\text{CoW}} + C_{\text{abort}}, \tag{17}$$

where C_{CoW} and C_{abort} are page-table operations that alias metadata and do not copy trunk KV. This expression is an accounting abstraction rather than a wall-clock predictor, since execution and transfer may overlap; we therefore measure end-to-end latency directly.

4.4 Pruning-Aware KV State

PD-Prune keeps one copy-on-write context tree per session (Appendix B.2, Figure 2). `fork` aliases the parent page table without copying the trunk KV; only residual pages are newly allocated. Shared pages are reclaimed only after their last live reference is released. A rejected residual allocates nothing; a probe that is not extended frees its residual and probe pages; abort frees the residual pages of every uncommitted candidate, $b\ell_i$ bytes each. On the next observation x, commit binds the child with the longest common prefix,

$$v^\star = \arg\max_v \text{LCP}(x, x_v), \qquad g_u \leftarrow g_u + 1, \tag{18}$$

publishes the full pages of $x_\star$ to the prefix index, and invalidates in-flight work tagged with an older generation g_u. Speculatively prefilled pages are never sampled from, so the emitted tokens equal those of a baseline that prefills committed tokens only. After commit the live occupancy is M_{spine}, and a pool of M_{pool} bytes holds $\lfloor M_{\text{pool}}/M_{\text{spine}} \rfloor$ sessions instead of $\lfloor M_{\text{pool}}/M_{\text{APC}} \rfloor$ while a prefix cache still retains aborted residuals.

Cross-step coordination. The committed spine of step t becomes the logical trunk of step $t{+}1$. Its KV residency on the prefill and decode instances is restored according to the underlying KV synchronization policy. A residual becomes a HIT only when its parent-aware prefix is already resident on the decode instance. Admission is re-evaluated at every step, so rejected branches never accumulate state across turns.

5 Evaluation

Setup. All GPU runs use vLLM on A100-SXM4 40GB GPUs with `bfloat16` weights, temperature 0, and APC enabled; hyperparameters, mix construction, and run-to-run variation are in Appendix C. Prefill and decode are physically disaggregated with a warm trunk (regime R_3), so prefill differences reflect residual admission only. Admission and probe results use Qwen3-8B on a *branching mix* of valid, repetitive, and malformed residuals for GSM8K and MATH-500; spine retention uses the full GSM8K test set on Qwen3-14B. We report computed prefill, wall-clock, peak KV, committed-answer accuracy, and any-finished coverage.

Prefill-Side Admission Complements Decode-Time Pruning. Table 2 (GSM8K block) places one admission policy in front of three decode-time pruners. Without admission, ESC, Speculative Rejection, and DPTS prefill the same 5008 tokens; they differ only in D^+ (131072, 98304, and 78336 tokens) because each reads generated text. Reject-only admission at $\tau_{\text{pre}} = 0.15$ rejects the four repetitive residuals before any kernel and reduces W to 3848 tokens (-23%) for every pruner, as Proposition 1 predicts. Committed accuracy is unchanged for every pruner and coverage remains $16/16$. The time reduction is largest for ESC (-12%), which would otherwise decode the repetitive residuals to the full budget, and small for pruners that already stop them early (-6%, -3%). Aggressive prefill-

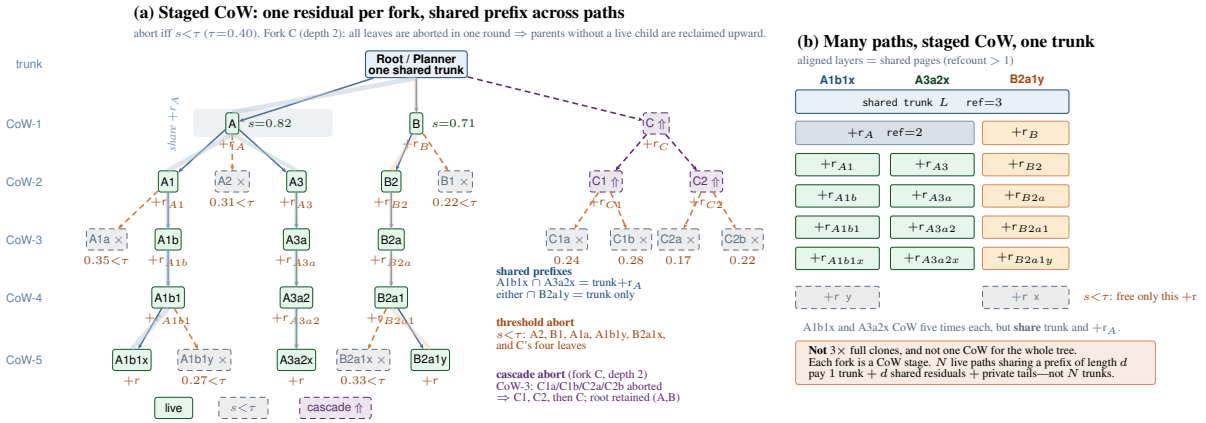


Figure 2: **Context tree.** (a) Each edge is one `fork`: the child aliases the parent and allocates only a residual (+r). Live paths share the trunk; A1b1x and A3a2x also share $+r_A$. All descendants of fork C are aborted at depth 2, so reclamation propagates up to C and stops at the root, which retains the live children A and B. (b) A shared layer is one physical page set with reference count greater than one; aborting a leaf frees only its residual.

Table 2: Main results on Qwen3-8B (branching mix, trunk warm). GSM8K: prefill-side admission composed with decode-side pruners. MATH-500: admission vs. probe under coverage. Settings in Appendix C.

Policy	admitted	W	time (s)	committed	any-finished
GSM8K; columns list ESC / Speculative Rejection / DPTS					
pruner only	16	5008	16.49 / 13.29 / 11.87	9 / 9 / 10	16 / 16 / 16
+ Reject-only	12	3848	14.45 / 12.53 / 11.57	9 / 9 / 10	16 / 16 / 16
+ Blanket	8	2768	10.95 / 10.95 / 10.96	9 / 9 / 9	16 / 16 / 16
MATH-500; out of 32					
full fan-out	16	12320	104.1	3	20
Blanket	8	6200	52.1	3	13
Reject-only	12	9332	71.4	2	16
Probe	8+4	28736	72.2	3	20

Table 3: ESC on GSM8K across fan-out width (Qwen3-8B, n=16). Each cell: committed / any-finished / seconds.

	k=4	k=8	k=16	k=32
ESC	9 / 16 / 8.62	8 / 16 / 10.90	9 / 16 / 16.48	9 / 16 / 32.78
+ Reject-only (0.15)	8 / 16 / 8.22	10 / 16 / 10.25	10 / 16 / 14.45	9 / 16 / 27.14
+ Blanket (0.45)	9 / 14 / 7.90	8 / 15 / 8.66	8 / 16 / 10.95	9 / 16 / 16.61

side rejection ("Blanket") at 0.45 also removes the malformed residuals and is faster (−34% for ESC); at k=16 coverage stays 16/16 and committed accuracy drops by at most one problem (DPTS). Table 3 shows that the trade-off depends on width: at k=4 and k=8, Blanket loses one to two problems of coverage, while reject-only stays within one of the pruner-only baseline. At k=32 the same reject-only gate cuts ESC wall-clock by 17% and blanket by 49%, both still at 16/16 coverage.

Decode-side probing mitigates the coverage loss of aggressive prefill-side rejection. On MATH-500 (Table 2, lower block), Blanket halves the time of the full fan-out but lowers coverage from 20/32 to 13/32: some residuals scored as malformed nevertheless lead to correct solutions once decoded. Reject-only admission reaches 16/32. At k=16, probing the uncertainty band with t_0=128 increases coverage to 20/32 at 72.2 s, 0.69× the full fan-out, and reduces D^+ from 524288 to 393216 tokens. Most uncertain residuals pass the decode-side refinement in this workload; Appendix D.5 provides the detailed probe outcomes and recomputation overhead. The cost of probing is visible in W: at k=16, evicted probe pages are recomputed on extension and W rises to 28736 tokens, which (14) accounts for as Δ_{ext}; at $k \le 8$ the overhead is below 300 tokens. The best probe length depends on width (Table 4): t_0=64 is preferable at k=8, and at k=4 no probe matches the full fan-out.

Connector volume. Table 5 instantiates Proposition 1 for 10 sessions with k=4, L=256, ℓ_i=32, and a cold trunk ($\bar{L}_{\mathcal{P}}$=$\bar{L}_{\mathcal{D}}$=L). Stock P/D is request-local, $(\nu^{\mathcal{P}}, \nu^{\mathcal{D}})$=$(k, k)$, so both columns equal $10\big(kL + \sum_i \ell_i\big)$=11520. Withholding the repetitive residual without aliasing leaves three request-local survivors, $(\nu^{\mathcal{P}}, \nu^{\mathcal{D}})$=$(3, 3)$, and both

Table 4: MATH-500 coverage (any-finished out of 32) and seconds across width. "Blanket" is aggressive prefill-side rejection (s_i<0.45); "Reject-only" rejects repetitive residuals; "Probe t_0" probes the band $[0.15, 0.45)$.

	k=4	k=8	k=16
Full fan-out	12, 25.3	15, 52.5	20, 104.1
Blanket	6, 19.1	11, 25.6	13, 52.1
Reject-only	11, 23.0	17, 41.5	16, 71.4
Probe t_0=64	8, 24.0	16, 42.0	18, 72.5
Probe t_0=128	9, 24.7	13, 43.3	20, 72.2

Table 5: Control-plane connector under P/D disaggregation (not GPU wall-clock). "P/D+rep." withholds repetitive residuals. Detailed rates and session parameters in Appendix C.

	APC	P/D	P/D+rep.	Alias+gate
Prefill W	11520	11520	8640	960
Connector X	0	11520	8640	960
Transfer	0	23.0 ms	17.3 **ms**	1.9 **ms**
Fan-out	138.2 ms	161.3 ms	121.0 **ms**	13.4 **ms**

columns fall to 8640. Of that 2880-token drop, $(4-3)\ L \times 10$=2560 is trunk-amplification reduction and 32×10=320 is the rejected residual: the two terms of (15). Alias+gate shares the trunk and, in this counter, ships only the miss tail of $\mathcal{K}$, $10 \times 3 \times 32$=960 tokens (0.08× stock P/D). That column charges X^{trunk}=0 rather than the single cold-trunk copy $\nu^{\mathcal{D}}$=1 would transfer, so the 8640→960 gap is the whole request-local trunk, $3L$ per session, i.e. the trunk-transfer term of (15) with nothing left to ship; the residual admission term does not change. The same ordering holds at $k \in \{8, 16, 32\}$ (Appendix D.6).

Pruning-aware spine retention lowers post-selection KV occupancy. On the full GSM8K test set (n=1319, Qwen3-14B, tp=2, k=4), retaining only the committed spine matches APC within one point of committed-answer accuracy (0.845 vs. 0.844) and reduces median peak KV usage to 0.72× that of APC (1030 vs. 1430 tokens; Appendix C.1). This result comes from state management alone (CoW-only, i.e., PD-Prune without admission), since admission does not change the committed spine. The ratio follows $(L + \ell_\star)/(L + \sum_i \ell_i)$ and is therefore governed by k and the trunk-to-residual ratio rather than by the model: it is 0.72–0.75× on GSM8K and 0.57–0.61× on Game of 24 across six families (Figure 11 in Appendix C.8), and it falls from about 0.90 at k=2 to 0.53–0.55 at k=8 (Appendix C.9). The copy-on-write fan-out incurs its own overhead: its median latency is 140 ms, compared with 87 ms for an APC hash hit.

Table 6: Three-step Tree-of-Thoughts (n=16, k=4, tp=2). Accuracy is committed-answer accuracy; e2e is seconds per problem; peak is spine tokens. The answer-marker stop in the PD-Prune column is orthogonal and is not attributed to PD-Prune.

	Qwen3-4B			Qwen3-8B		
	APC	CoW-only	PD-Prune	APC	CoW-only	PD-Prune
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

Multi-step agent sessions. We run a three-step Tree-of-Thoughts session (k=4 per step, n=16, Qwen3-4B and Qwen3-8B, tp=2) in which each step's spine is the next step's trunk (Table 6). Admission runs before every fan-out and prefills 48 of the 192 residuals. Peak KV is 0.86× APC on GSM8K and 0.82× on Game of 24 for both CoW-only and PD-Prune. Committed accuracy matches APC on Qwen3-4B GSM8K, is one problem lower on Qwen3-8B GSM8K (15 vs. 16 of 16), and is one to three problems higher on Game of 24. The answer-marker stop in the PD-Prune column is an orthogonal optimization and is not attributed to PD-Prune. End-to-end times in Table 6 therefore mix admission with this stop, and we do not treat those reductions as a PD-Prune result.

6 Conclusion

We presented PD-Prune, a cross-stage pruning framework for disaggregated prefill–decode agentic serving. On Qwen3 models and reasoning workloads, prefill-side admission reduces computed prefill by 23% at matched committed-answer accuracy, a 128-token probe restores MATH-500 coverage to 20/32 at 0.69× the full-fan-out wall-clock, and spine retention reduces median peak KV usage to 0.72× that of APC. These results show the benefit of filtering branches at the earliest informed stage and of retaining only the committed spine.

Limitations. PD-Prune currently relies on lightweight heuristic scores for admission and probing, evaluated primarily on reasoning-oriented branching workloads. Although the framework is agnostic to the scoring function, the separation observed in our experiments may not generalize directly to heterogeneous agent traces, tool-use tra-

jectories, or less structured residuals. Evaluating learned or workload-adaptive scoring policies on large-scale production traces remains an important direction for future work.

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A Background

A.1 Branching in Agentic and Reasoning Workloads

Chain-of-thought produces one trace [32]; Tree-of-Thoughts, self-consistency, and Monte Carlo tree search produce many and then select [35, 31, 39]. In the Game-of-24 setting of Tree-of-Thoughts, a proposal prompt emits candidate equations, a

value prompt labels each candidate, and breadth-first search retains a beam [35]. Branching controllers follow the same pattern: a coding or tool-using agent proposes several continuations of one context, executes or observes one, and discards the rest [36, 34, 30]. On GSM8K and MATH the accuracy gain of such width is well documented [8, 12, 19], and so is its token cost [21, 4].

A.2 Decode-Time Path Pruning

Early-stopping self-consistency draws a window of samples and stops once the window agrees [17, 31]. Speculative Rejection ranks partial generations at a reward horizon and halts the lower quantile [26]. DPTS generates a mini-step on every active node and early-stops low-confidence children [10]. STOP organizes these signals by source and learnability and trains an internal token that predicts futility from a short prefix [4]; EquivPruner merges semantically equivalent actions [21]. Each signal is a function of generated text. On our GSM8K mix at k=16, ESC, Speculative Rejection, and DPTS therefore prefill the same 5008 tokens and differ only in decode volume (131072, 98304, and 78336 tokens).

A.3 Prefix Caching and Disaggregated Transfer

APC and RadixAttention index full blocks by chained keys, $h_j = \mathrm{hash}(h_{j-1}, \mathrm{block}_j, \mathrm{extra})$ [14, 41]. A request reuses a block only if the block and its entire parent chain are resident. Whether k siblings issued together share the trunk depends on the cache regime of §3.2: a resident trunk (R3) is shared by all siblings; a cold trunk is computed once if siblings are serialized behind it (R2), and between one and k times if they are admitted concurrently, depending on how the scheduler treats blocks that are still being computed (R1). Connector multiplicity $\nu_u^{\mathcal{D}}$ is independent of that prefill multiplicity: request-local transfer can move the trunk k times even when $\mathcal{P}$ materializes it once. APC thus addresses trunk reuse, but it neither rejects a residual nor decides which pages cross a connector.

PD disaggregation places prefill and decode on separate instances and transfers KV between them [42, 25]. Every admitted request is prefilled on $\mathcal{P}$, and the pages that $\mathcal{D}$ lacks are transferred; a residual that the branching controller later discards has already consumed both. Program- and workflow-level schedulers raise the scheduling unit above the request [23, 13, 16, 33], and KV retention removes trunk recomputation across tool gaps; the remaining per-step cost is the residual suffixes of siblings that the branching controller can enumerate before the observation returns.

Table 7: Notation.

Symbol	Domain	Meaning
$L, \ell_i, \ell_\star$	$\mathbb{N}$	trunk; residual i; committed residual
$x_u, \rho_i, x_\star$	tokens	trunk; residual; spine
$\mathcal{P}, \mathcal{D}$	—	prefill and decode instances
$c_u^{\mathcal{X}}, m_i^{\mathcal{X}}$	tokens	resident prefix of trunk and residual on $\mathcal{X}$
s_i, s_i'	$[0,1]$	residual score; probe score
$\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}}$	$[0,1]$	rejection and extension thresholds
a_i	$\mathcal{A}$	HIT, ADMIT, PROBE, REJECT(Definition 2)
e_i	$\{0,1\}$	probe of i is extended
t_0, D, d_i	tokens	probe length; budget; decode tokens of i
W_i, X_i	tokens	computed prefill; connector tokens
$\bar{L}_{\mathcal{X}}$	tokens	trunk tokens missing on $\mathcal{X}$
$\nu_u^{\mathcal{X}}$	$\{0,\dots,k\}$	trunk multiplicity on $\mathcal{X} \in \{\mathcal{P}, \mathcal{D}\}$
σ_u	$\{0,1\}$	publication barrier; 1 serializes siblings
Γ_u	—	page table of node u
$\mathcal{R}_u$	—	regime $(\nu_u^{\mathcal{P}}, \nu_u^{\mathcal{D}}, \sigma_u)$
W_i^{ext}	tokens	recomputation on probe extension
$\mathcal{R}, \mathcal{H}, \mathcal{E}$	$\subseteq [k]$	rejected, hit, and extended residuals
b, P	bytes, tokens	KV bytes per token; page size
M_{pool}	bytes	KV pool capacity
$B_t = B_t^c + B_t^s$	tokens	committed and speculative chunk budget

A.4 Prefill, Decode, and Idle Time

Prefill processes the prompt in parallel and is compute-bound; decode emits one token per step and is memory-bandwidth-bound [37, 42, 25]. PagedAttention stores KV in non-contiguous pages [14], and chunked prefill co-schedules a prompt chunk with a decode batch [2]. For agents, the prompt of step $t{+}1$ extends that of step t after a tool interval of variable length [16, 33], during which known suffixes can be prefilled (Appendix B.5). Figure 3 shows where each existing mechanism can first discard a sibling, and Table 7 summarizes the notation.

B System Details

PD-Prune sits between the branching controller and the LLM engine (Figure 1). The branching controller exposes branch structure, PD-Prune decides and binds, and the engine executes prefill, transfer, and decode.

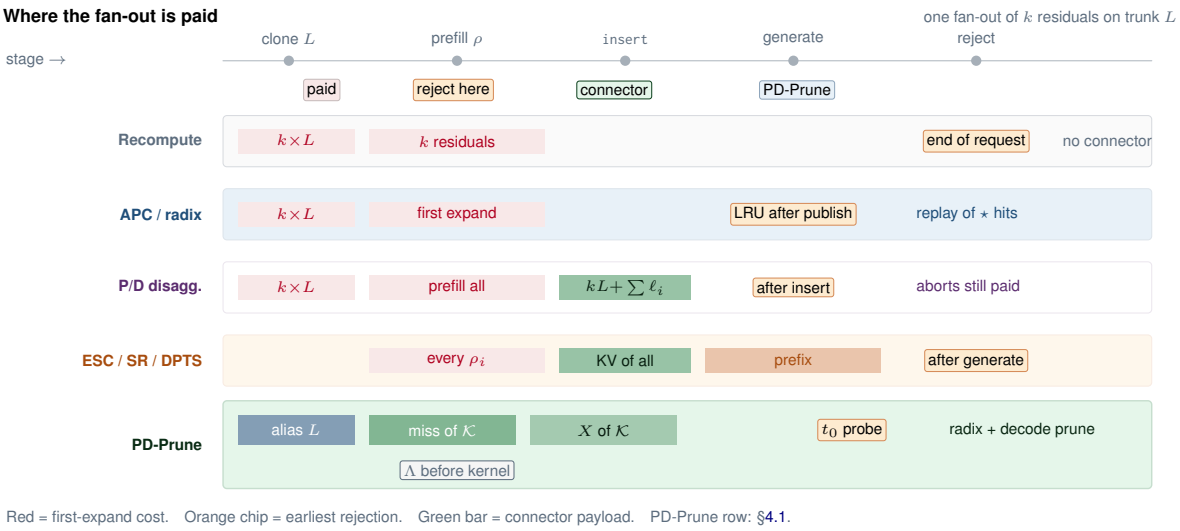


Figure 3: **Where each mechanism acts on one fan-out of k residuals over a trunk of length L.** Red bars denote prefill issued before any sibling is discarded, orange chips the earliest rejection point, and green bars connector volume. Trunk volumes for the prefix-cache rows assume the cold concurrent regime R1; in regimes R2–R4 the trunk is computed at most once (§3.2). The PD-Prune row follows §4.1.

B.1 Interface

`open(session)`	Create a root; prefill the initial context.
`fork(u, bid, tok?)`	Child of u, optionally with a known suffix.
`speculate(v, pri)`	Mark v eligible for idle-time prefill.
`commit(u, tokens)`	Bind by LCP ((18)); start decode.
`join(children, pol)`	Merge a fan-out (`all`/`first`/`kofn`).
`abort(v)` / `close`	Drop a subtree; end the session.

Only `commit` lies on the critical path of user-visible tokens.

B.2 Copy-on-Write Context Tree

Definition 3 (Context tree)**.** A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u with $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|{:}]$, a mode in $\{\textsc{Spec}, \textsc{Commit}, \textsc{Idle}, \textsc{Dead}\}$, a generation g_u, and a page table $\Gamma_u = (\text{parent}, \text{alias_len}, \text{pages}_u)$.

The tree maintains four invariants: (i) prefix closure; (ii) at most one committed child per node outside a join; (iii) speculative nodes are never sampled from; (iv) a physical page remains live while at least one non-dead node maps it. `fork`$(u, \cdot)$ creates a metadata alias to the parent page table and registers the child as a live reference; no trunk KV data are copied, and only residual pages are newly allocated. A write to a shared or read-only page copies only its valid rows. Shared pages are reclaimed only after their last live reference is released. A multi-level fan-out is a sequence of such forks (Figure 2); abort frees residual pages only, and a subtree without live descendants is reclaimed up to the nearest ancestor with a live child. Commit publishes the full pages of $x_\star$ to the prefix index so that a later request whose parent-aware prefix is already resident on $\mathcal{D}$ can take HIT. Eviction is node-granular: speculative leaves first, then idle committed leaves, then the spine [16, 33].

B.3 Speculative Prefill of Known Suffixes

A candidate of parent u is a pair $(x^{\mathrm{k}}, \hat{x}^{\mathrm{o}})$ of a known suffix and a predicted observation, with admission probability p and expected prefix-hit rate q. Known suffixes ($p{=}q{=}1$) are scheduled before predicted observations ($q{<}1$), subject to the idle time $T_{\text{idle}}(u)$, the available KV capacity, and a branch cap. Unlike speculative decoding [15, 6], the object prefilled is a future *prompt* prefix rather than draft output tokens. On commit, the pages covered by $\text{LCP}(x, x_{v^\star})$ become committed, the unmatched tail is prefilled, every sibling is aborted, and jobs tagged with a stale generation are dropped by generation tag.

B.4 Admission Details

Scorers. $\text{Score}_{\text{pre}}$ assigns 0.02 to a residual containing a substring of 8–40 characters repeated at least four times, 0.22 to a residual containing malformed mathematical tokens (e.g., runs of asterisks or `nan`), and otherwise a weighted combination of alphanumeric fraction and character diversity,

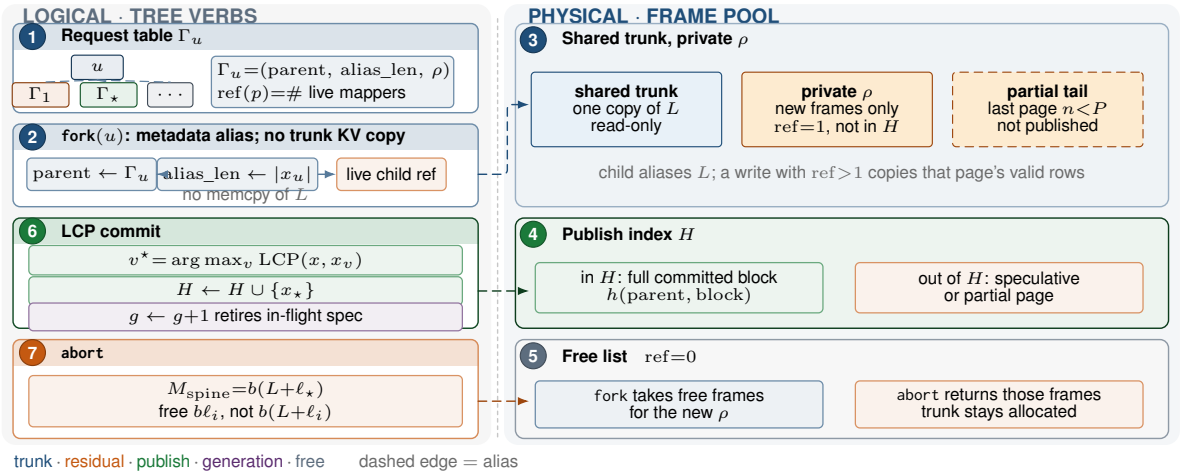


Figure 4: **Context-tree page pool** (pruning-aware KV state). `fork` aliases the parent page table without copying the trunk KV and allocates only residual pages ①–③. LCP commit publishes full blocks of $x_\star$ into H so a later request whose parent-aware prefix is already resident on $\mathcal{D}$ may take HIT ④⑥. `abort` returns residual pages only; the live occupancy is $M_{\text{spine}}{=}b(L{+}\ell_\star)$ ⑤⑦.

which is close to 1 for natural prose. $\text{Score}_{\text{dec}}$ applies the same repetition and malformation tests to the generated prefix (0.04 and 0.16), assigns 0.92 once a finished answer marker (#### followed by a number, or \boxed{) appears, and otherwise uses a weighted combination that also rewards digits and operators. Both scorers run on the CPU and do not share parameters.

Ordering of the action. (8) reads decode-side residency first, because a residual already resident on $\mathcal{D}$ costs neither prefill nor transfer. It then protects index 0, so the top-ranked continuation is always served. Only then does it consult the score. A page resident only on $\mathcal{P}$ reduces W_i but not X_i; the residual is admitted with $W_i{=}\ell_i - m_i^{\mathcal{P}}$ and $X_i{=}\ell_i - m_i^{\mathcal{D}}$.

Spans. Table 8 summarizes prefill, transfer, and post-abort memory per fan-out for a cold trunk.

B.5 Critical-Path Placement

After abort the live occupancy is M_{spine}, so a pool of M_{pool} bytes holds

$$C = \left\lfloor M_{\text{pool}}/M \right\rfloor \tag{19}$$

sessions, with $M{=}M_{\text{spine}}$ for PD-Prune and $M{=}M_{\text{APC}}$ while a prefix cache retains aborted residuals. On tick t the chunk budget [2] is split by the committed rate $\underline{s}_r$ [40],

$$B_t^{\text{c}} = \min\Big(B_t,\ \sum_{r\in R_t^{\text{c}}} \max(\underline{s}_r\Delta,\ 1)\Big), \tag{20}$$

$$B_t^{\text{s}} = B_t - B_t^{\text{c}}. \tag{21}$$

B_t^{c} decodes committed spines; B_t^{s} prefills admitted residuals or known suffixes ($p{=}1$) when $T_{\text{pre}}(|w|) \le T_{\text{idle}}(u)$. Speculatively prefilled tokens are never sampled from.

B.6 Multi-Step Pipeline

Algorithm 2 shows how Alg. 1 repeats across agent steps on the PD path: admit on the controller, prefill on $\mathcal{P}$, ship on the connector, then probe and decode on $\mathcal{D}$. The committed spine of step t becomes the logical trunk of step $t{+}1$. Its KV residency on $\mathcal{P}$ and $\mathcal{D}$ is restored according to the underlying KV synchronization policy.

C Additional Evaluation

This appendix reports the full experimental setup, the GSM8K forest on Qwen3-14B (§C.1), details of the admission and probe experiments (§C.2–§C.5), a Qwen3-8B forest with an answer-marker decode stop (§C.6), the multi-step sessions (§C.7), cross-model results (§C.8), and sensitivity to width, depth, and trace length (§C.9).

Common setup. All GPU runs use vLLM on A100-SXM4 40GB GPUs with `bfloat16` weights, temperature 0, and APC enabled. Prefill and decode are physically disaggregated: $\mathcal{P}$ and $\mathcal{D}$ run

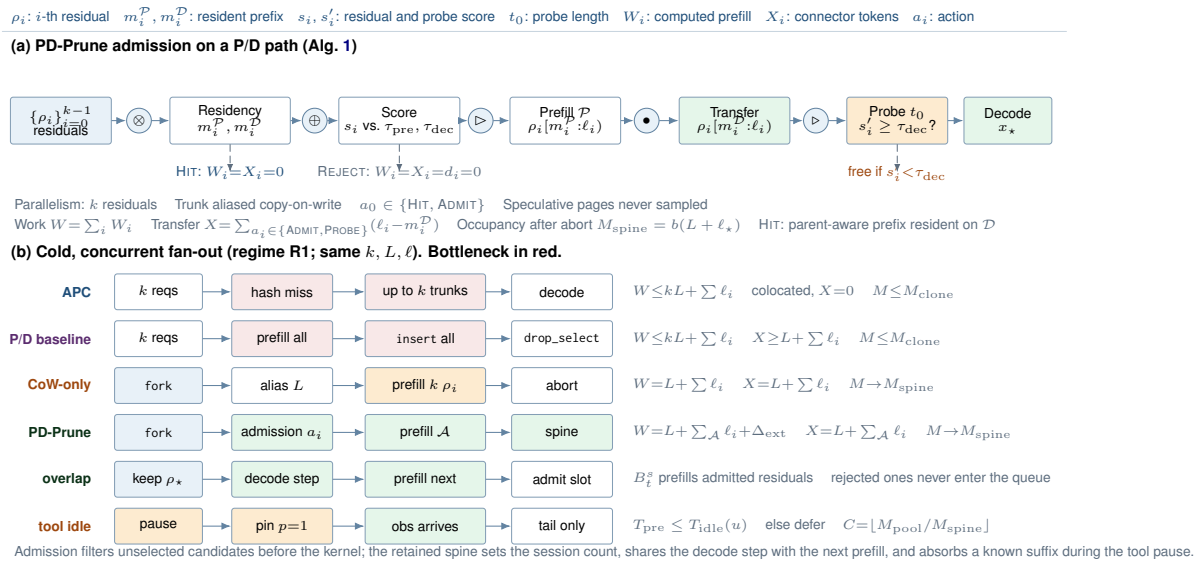


Figure 5: **Admission on the request path.** (a) The action of (8) is evaluated before `insert`; only admitted and probed residuals reach the prefill kernel. (b) Prefill and connector volume of a cold, concurrent fan-out (regime R1) for APC, a baseline PD deployment, the copy-on-write tree without admission (CoW-only), and PD-Prune. The overlap row prefills the next residual during decode; the tool-interval row prefills a known suffix while $T_{\text{pre}} \leq T_{\text{idle}}$.

Table 8: Per-fan-out spans with a cold trunk ($c_u^{\mathcal{P}}{=}c_u^{\mathcal{D}}{=}0$) and no resident residual pages. The PD baseline is request-local (R_1, $\nu^{\mathcal{P}}{=}\nu^{\mathcal{D}}{=}k$); serialized admission (R_2) and explicit fork (R_4) set the trunk multiplicity to 1, so the trunk term becomes L. $\mathcal{A}$ is the set of admitted and probed residuals; T_{ab} is the abort cost in pages; "replay" is a later request for the committed spine.

	W	X	M after abort	T_{ab}	trunk copies	replay W
Recompute	$kL+\sum_i \ell_i$	0	$kL+\sum_i \ell_i$	$\Theta(kL/P)$	k	$L+\ell_\star$
APC (R1)	$\leq kL+\sum_i \ell_i$	0	$\leq kL+\sum_i \ell_i$	LRU	1–k	0
PD baseline (R1)	$\leq kL+\sum_i \ell_i$	$\geq kL+\sum_i \ell_i$	$\leq kL+\sum_i \ell_i$	LRU	1–k	0
CoW-only	$L+\sum_i \ell_i$	$L+\sum_i \ell_i$	$L+\ell_\star$	$\Theta(\ell_i/P)$	1	0
PD-Prune	$L+\sum_{i\in\mathcal{A}} \ell_i+\Delta_{\text{ext}}$	$L+\sum_{i\in\mathcal{A}} \ell_i$	$L+\ell_\star$	$\Theta(\ell_i/P)$	1	0

Algorithm 2 PD-Prune multi-step pipeline. Each step invokes Alg. 1.

Require: initial trunk x_u; thresholds $(\tau_{\text{pre}}, \tau_{\text{dec}}, t_0, D)$
Ensure: committed spines $\{x_\star^{(t)}\}$
OpenTree(x_u)
for each agent step t **do**
  $\{\rho_i\} \leftarrow \text{Expand}(x_u; k)$ ▷ branching-controller fan-out
  Controller: run Alg. 1 $\to x_\star^{(t)}$, (W, X, D^+)
  $\mathcal{P}$**:** prefill only $\mathcal{P}$-miss pages of admitted / probed residuals
  Connector: ship only $\mathcal{D}$-miss pages
  $\mathcal{D}$**:** probe $\to$ decode-time prune $\to$ decode survivors
  $x_u \leftarrow x_\star^{(t)}$ ▷ spine becomes next logical trunk
end for
return $\{x_\star^{(t)}\}$

as separate vLLM instances connected by a KV connector, and admitted residuals reach $\mathcal{D}$ only through that connector. We measure computed prefill W, decode tokens D^+, peak KV, and end-to-end wall-clock. The trunk is warmed before every fan-out (regime R3), so prefill differences reflect residual admission only. Models include Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B (R1-8B).

Admission and probe workloads. The admission experiments use Qwen3-8B at tp=1 on a *branching mix*: for each problem, half of the k residuals are distinct solution strategies (valid), a quarter are repetitive (repetition loops), and a quarter are syntactically malformed. GSM8K uses $n{=}16$ problems and $D{=}512$; MATH-500 uses the first 32 problems of level ≥ 4 and $D{=}1024$. On GSM8K, "Reject-only" rejects $s_i{<}0.15$ and admits the rest; "Blanket" rejects $s_i{<}0.45$. On MATH-500 (no decode-time pruner), "Probe" uses $(\tau_{\text{pre}}, \tau_{\text{dec}}, t_0) = (0.15, 0.45, 128)$, and its W includes W^{ext} from evicted probe pages; the committed residual is fixed to index 0, so committed accuracy is insensitive to admission by construction and coverage is the informative metric. Prefill

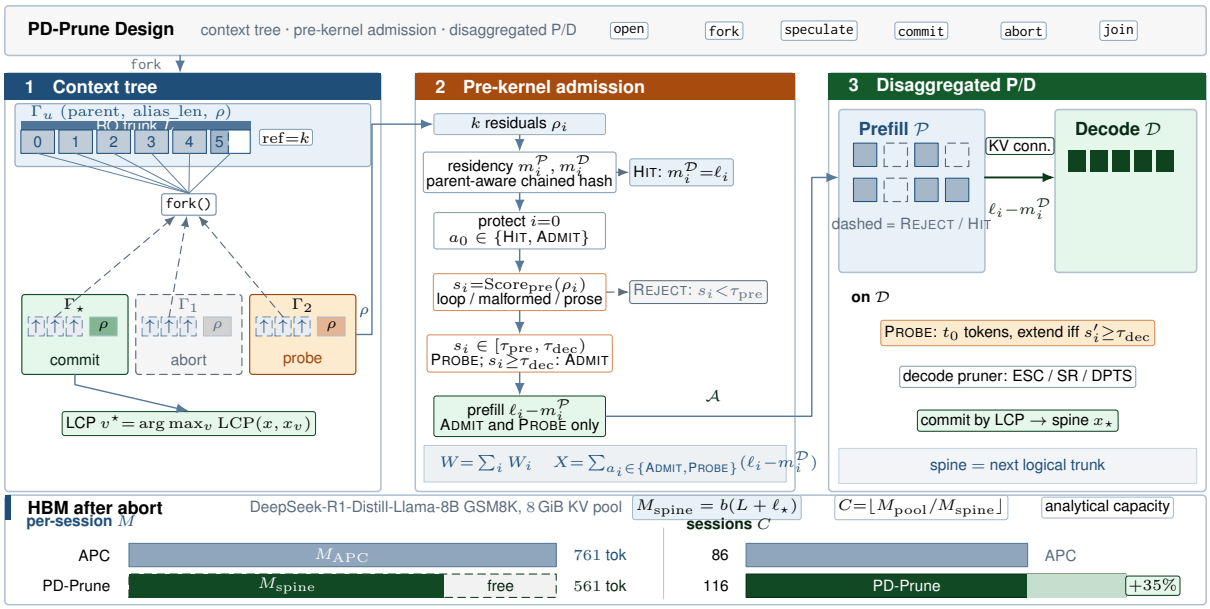


Figure 6: **PD-Prune design.** (1) CoW state: Γ_u aliases one trunk, `fork` copies no trunk KV, and LCP commit publishes $x_\star$. (2) Controller policy π: observation $o_i{=}(m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, s_i)$ maps to $a_i \in \{\text{HIT}, \text{ADMIT}, \text{PROBE}, \text{REJECT}\}$; only $\mathcal{A}$ enters the GPU path. (3) Disaggregated execution: $\mathcal{P}$ prefills the miss of $\mathcal{A}$, the connector ships decoder-missing pages, and $\mathcal{D}$ runs the t_0 probe then the decode-time pruner before LCP commit. Bottom: M_{spine} frees HBM and raises analytical session capacity.

W is identical across decode-time pruners because those pruners act after prefill.

Metrics. Our primary metric is *committed-answer accuracy*: the answer of the candidate the branching controller commits (index 0 for MATH-500, the pruner's selection for GSM8K) matches the reference. We also report *any-finished* coverage, i.e., whether some full-budget branch is correct, which upper-bounds what a better selector could commit. An answer is correct when its final number matches the reference. With $n{=}16$, identical ESC configurations at $k{=}16$ commit between $8/16$ and $11/16$ across runs (Appendix D.2), so differences of up to three problems are within run-to-run variation. Tree-of-Thoughts forests elsewhere use $k{=}4$ and $D{=}512$ (2048 for contest problems) in batches of eight.

Connector microbenchmark. Table 5 uses a control-plane accounting over 10 sessions with $k{=}4$, $L{=}256$, and $\ell_i{=}32$, charging prefill at $12\,\mu\text{s/token}$ and transfer at $2\,\mu\text{s/token}$. "P/D+rep." is $\tau_{\text{pre}}{=}0.15$ on a stock (non-aliased) prefiller and withholds repetitive residuals; "Alias+gate" aliases the trunk and ships only the miss of $\mathcal{K}$. These numbers are not GPU wall-clock.

Table 9: GSM8K test, Qwen3-14B, tp=2, $n{=}1319$; medians over 165 batches of eight.

	recompute	APC	CoW-only
Committed accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	2.89×	1.00×	0.72×
Fan-out, median (ms)	326	87	140

C.1 GSM8K Forest on Qwen3-14B

The full GSM8K test set ($n{=}1319$) is run on Qwen3-14B at tp=2 in 165 batches of eight (Table 9). This experiment isolates state management: no residual is rejected, so the copy-on-write tree is compared with recomputation and APC on the same committed path.

Median peak KV is $0.72\times$ APC and $0.25\times$ recomputation, and accuracy is within 0.3 points of both. Fan-out latency does not follow the same order: the copy-on-write fan-out takes 140 ms against 87 ms for APC, because fork and abort run on the expand path while an APC hit does not. The same peak ordering (0.58–$0.80\times$ APC) holds on seven math datasets for Qwen3-14B, Qwen3-8B, and R1-8B at tp=2 and 4, and peak KV does not change with tensor-parallel width, consistent with (1).

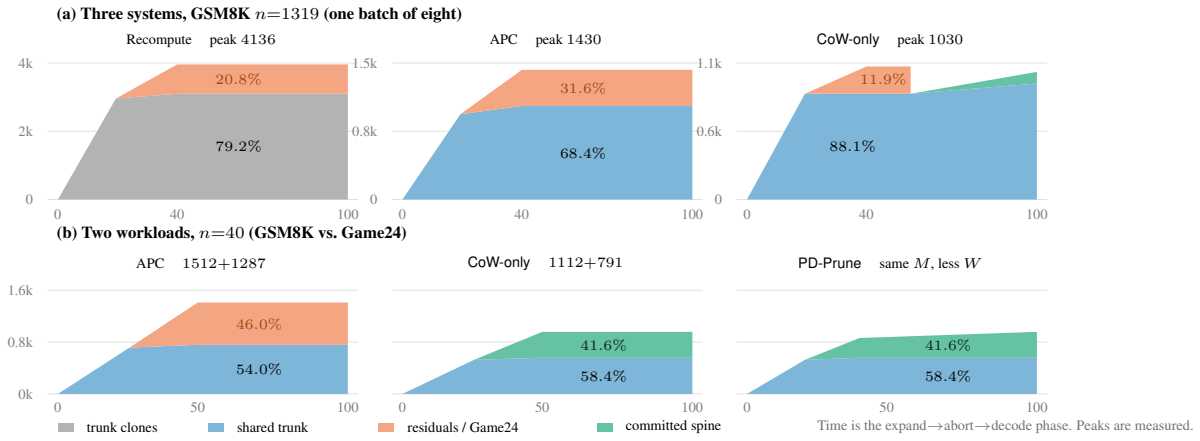


Figure 7: **Stacked KV occupancy** over expand, abort, and decode. (a) One GSM8K batch of eight on Qwen3-14B (per-panel y-scale; peaks 4136, 1430, 1030). (b) GSM8K (bottom) and Game of 24 (top), n=40, for APC, CoW-only, and PD-Prune; CoW-only and PD-Prune share M, and PD-Prune computes fewer residual tokens.

C.2 Probe Admission on MATH-500

The MATH-500 experiments use Qwen3-8B at tp=1, the first 32 problems of level 4 or 5, and D=1024. Half of the k residuals are contest strategies; the other half alternate repetitive and malformed text. Coverage counts a problem when some branch that ran all 1024 tokens is correct; the committed residual is index 0. Aggressive prefill-side rejection ("Blanket") lowers coverage from 12/32 to 6/32 at k=4 and from 20/32 to 13/32 at k=16, whereas no repetitive residual yields a correct answer at any k, so rejecting repetitive residuals does not reduce coverage. At k=16 a 128-token probe increases coverage to 20/32 in 72.2 s instead of 104.1 s. Most uncertain residuals pass the decode-side refinement in this mix; probing the repetitive residuals as well extends none of them and costs 74.4 s. At k=8, t_0=64 reaches 16/32 in 42.0 s, exceeding the full fan-out (15/32 in 52.5 s); at k=4 the t_0=64 probe reaches 8/32 in 24.0 s, and reject-only admission (11/32 in 23.0 s) is the closest non-probe setting. Figure 8 plots Table 4.

C.3 Admission on GSM8K

The GSM8K admission experiments use the branching mix with n=16 and D=512 on Qwen3-8B at tp=1. Because the heuristic scores take three values on this mix (repetitive 0.02, malformed 0.22, valid $\approx$ 1), any $\tau_{\text{pre}} \in (0.02, 0.22]$ yields the same admission set as 0.15, and any $\tau_{\text{pre}} \in (0.22, 1)$ yields the same set as 0.45; a denser threshold sweep therefore adds no information on this workload. Figure 9 plots ESC across widths and the three pruners at k=16. At k=8, reject-only admission keeps ESC at 16/16 coverage (10.84 to 10.19 s) and raises Speculative Rejection and DPTS from 15/16 to 16/16.

C.4 Decode-Only, Prefill-Only, and Combined Pruning

Table 16 (Appendix D.3) factorizes the k=16 results. Decode-only pruning leaves W at 5008 tokens; prefill-only admission decodes every admitted residual to D; the combination retains the prefill reduction and adds the pruner's decode-time termination. At τ_{pre}=0.45 the pruners terminate no additional residuals, because every residual they would terminate has already been rejected.

C.5 Score-Based versus Top-1-Only Admission

Admitting only index 0 is a different policy from scoring every residual. At k=16 and τ=0.45, Top-1-only admission prefills 414 tokens but lowers coverage to 11/16 under ESC and 9/16 under Speculative Rejection, while score-based admission of the eight residuals above τ prefills 2768 tokens and retains 16/16 (Table 17). Committed accuracy is similar for both policies (9–11 of 16), so the difference lies in the coverage available to a better selector.

Capacity from measured peaks. For R1-8B on GSM8K (n=16, k=4), measured median peaks are 761 tokens under APC and 561 under the context tree. With 32 layers, 8 KV heads of dimension 128, and `bfloat16`, one token occupies 131072 bytes, so an 8 GiB KV budget holds 65536 tokens, i.e., 86 APC trees or 116 spines by (19) (+35%).

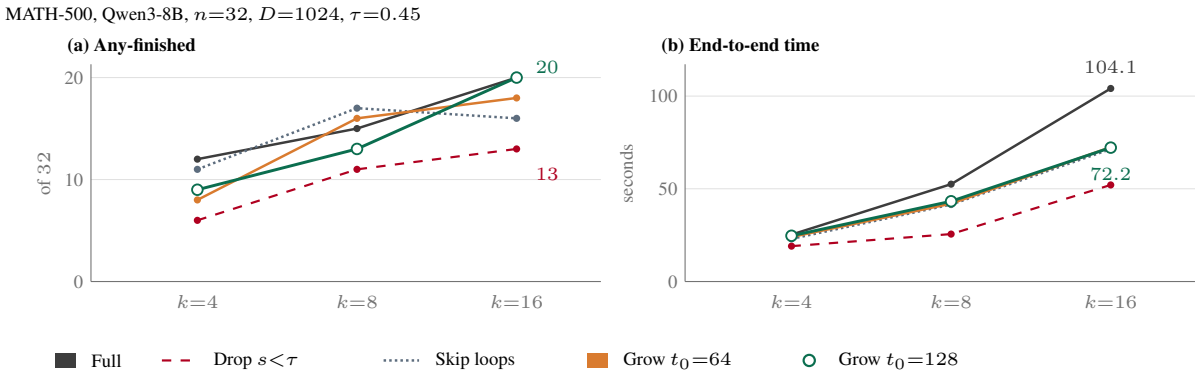


Figure 8: **Probe admission on MATH-500** (Table 4). Blanket rejection is fast but loses coverage; a t_0=128 probe increases coverage to 20/32 at k=16 in 72.2 s instead of 104.1 s.

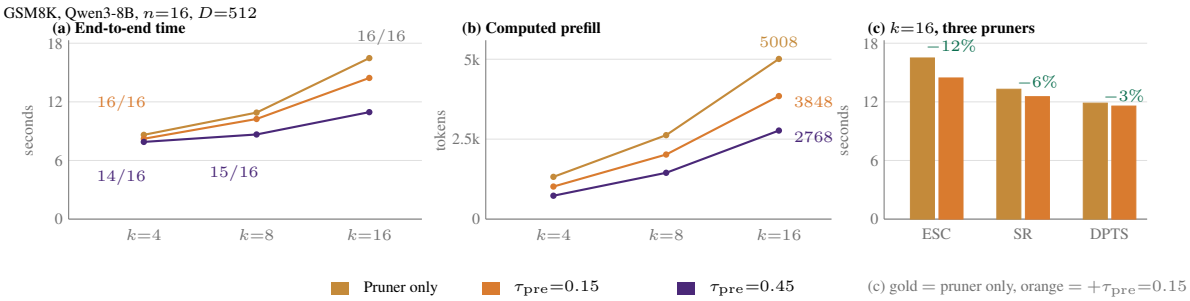


Figure 9: **Admission threshold on GSM8K** (Tables 2 and 3). (a) ESC wall-clock across k, labeled with any-finished coverage. (b) Computed prefill: −23% at τ_{pre}=0.15 and −45% at 0.45. (c) Reject-only admission in front of each pruner at k=16.

This is an analytical capacity bound; we do not report SLO-constrained throughput, which would additionally depend on scheduling and batching.

C.6 Qwen3-8B Forest with an Answer-Marker Stop

This experiment runs Qwen3-8B on GSM8K with n=80 and D=512 at tp=2 and 4; Game of 24 uses Tree-of-Thoughts with k=4, and HumanEval uses a ReAct wrap-and-recover loop with k=2. The PD-Prune runs additionally stop committed decode when the last non-empty line contains a finished answer, which reduces D^+ to about one third. The answer-marker stop is an orthogonal optimization and is not attributed to PD-Prune. On this trace the scorer admits every issued residual, so the end-to-end ratio in Table 10 reflects the decode stop rather than admission.

Peak KV is 0.75× APC on GSM8K, 0.62× on Game of 24, and 0.96× on HumanEval, where k=2 makes the residual a small fraction of the context. Accuracy is 1–3 points below APC (0.863 vs. 0.900 at tp=2; 0.925 vs. 0.938 at tp=4), which we attribute to the decode stop truncating a few answers.

Table 10: Qwen3-8B forest, n=80, D=512. PD-Prune includes an answer-marker decode stop. Peak KV does not depend on tp; G24 and HE are the Game of 24 and HumanEval peaks.

	peak	D^+	e2e (s)	acc.	e2e/APC	G24	HE
APC, tp=2	1582	40960	36.0	0.900	—	1289	1970
PD-Prune, tp=2	1182	13356	28.2	0.863	0.78	793	1882
APC, tp=4	1582	40960	22.7	0.938	—	1289	1970
PD-Prune, tp=4	1182	12826	16.9	0.925	0.75	793	1882

C.7 Multi-Step Tree-of-Thoughts

Each problem is one session of k=4 candidates over three steps on Qwen3-4B and Qwen3-8B (tp=2, n=16), issued as its own request. Steps 1 and 2 decode a 32-token step, abort the non-selected candidates, and promote the selected spine to the next trunk; step 3 decodes the final answer with D=512 (GSM8K) or D=256 (Game of 24). On Game of 24 the steps propose an equation, continue on the remaining cards, and write the final expression; on GSM8K they set up the algebra, revise the previous step, and state the final number. Admission runs before every fan-out, and the 32-token

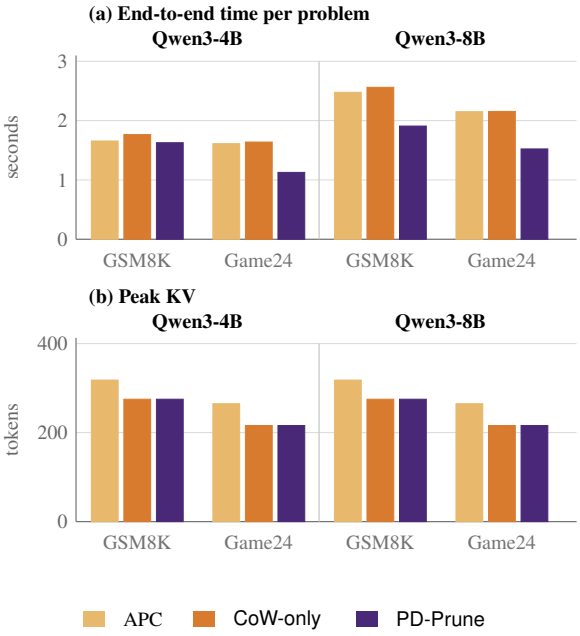


Figure 10: **Three-step Tree-of-Thoughts** (n=16, tp=2). (a) End-to-end seconds per problem; the answer-marker stop in PD-Prune is orthogonal and is not attributed to PD-Prune. (b) Peak KV of the committed spine.

step is decoded only for admitted residuals. Per-phase timings are not comparable across systems, because the APC baseline attributes step decoding to the expansion phase whereas ours attributes it to decode; Table 6 therefore reports end-to-end time only.

C.8 Cross-Model Results

The forest is repeated on six families (n=40, D=512, tp=2 and 4). APC retains the shared trunk and all residuals, $M_{\text{CoW}} = b(L + \sum_i \ell_i)$, while the context tree retains $M_{\text{spine}} = b(L + \ell_\star)$; their ratio $(L + \ell_\star)/(L + \sum_i \ell_i)$ is independent of the model and of tensor-parallel width. For similar residual lengths it reduces to $(L + \ell)/(L + k\ell)$, which decreases as L/ℓ decreases: 0.72–0.75 on GSM8K with a long trunk and 0.57–0.61 on Game of 24 with a short one (Figure 11). Relative to recomputation, which stores M_{clone}, the spine is 0.25×.

Figure 12 repeats the comparison at n=80; Figure 13 shows per-item decode lengths.

Table 11 isolates the answer-marker stop on three further models (n=32). The stop reduces D^+ by 23–69% and leaves the spine peak unchanged, but its effect on accuracy is model-dependent: +2 problems on R1-8B, −1 on Llama-3-8B, and −5 on Mistral-7B (0.344 to 0.188), where the model emits intermediate answer markers before finishing. The stop is therefore not a safe default, is not part of the admission results in §5, and is not attributed to PD-Prune.

Table 11: Spine peak and an optional answer-marker decode stop (n=32, tp=2; GSM8K D=512, Game of 24 D=256). The PD-Prune column applies both the context tree and the stop.

Model	Metric	APC	PD-Prune
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
Llama-3-8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
Mistral-7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

Table 12: GSM8K peak KV (tokens) versus branching (n=16, tp=2, D=512). PD-Prune has the same peak as CoW-only.

Model		k=2	k=4	k=8	depth 2
Llama-3-8B	APC	610	750	998	814
	CoW-only	550	550	550	614
	ratio	0.90	0.73	0.55	0.75
Mistral-7B	APC	629	769	1033	841
	CoW-only	569	569	569	641
	ratio	0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	CoW-only	561	561	561	641
	ratio	0.89	0.74	0.53	0.76

C.9 Width, Depth, and Long Traces

GSM8K runs use n=16, D=512, and tp=2 on Llama-3-8B, Mistral-7B, and R1-8B, varying k or adding a second Tree-of-Thoughts level (Table 12 and Figure 14). The spine peak is invariant in k, while the APC peak grows with the union of residuals, so the ratio falls from 0.89–0.90 at k=2 to 0.53–0.55 at k=8; at depth 2 it is 0.75–0.76. Admission leaves the peak unchanged; at k=8 it rejects 28 branches and reduces fan-out time relative to CoW-only from 307 to 179 ms on Llama-3-8B (APC: 224 ms), 349 to 191 ms on Mistral-7B, and 363 to 194 ms on R1-8B. With a batch of 16, the ratio is 0.72–0.73.

Table 13 reports contest problems with D=2048 (MATH-500 n=200, AIME n=73). The peak ratio is 0.78–0.91, and accuracy is within 3 points of APC on MATH-500 and within one problem

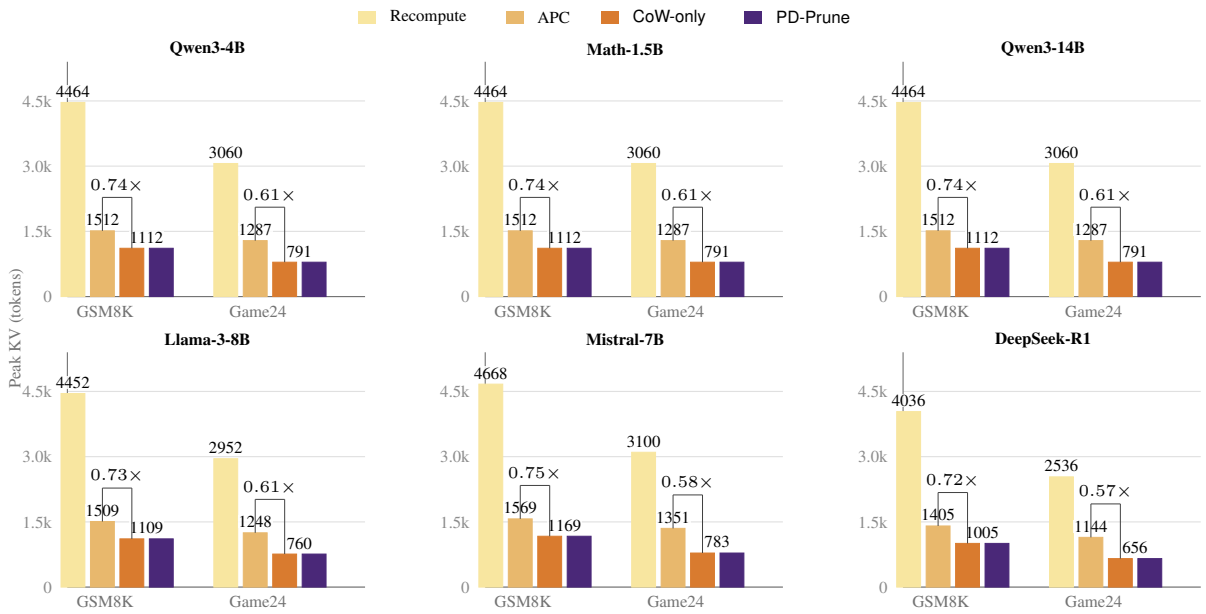


Figure 11: **Peak KV across six model families** (n=40, tp=2). Bars are recompute, APC, CoW-only, and PD-Prune; CoW-only and PD-Prune retain the same spine. The annotated ratio is $M_{\text{spine}}/M_{\text{APC}}$.

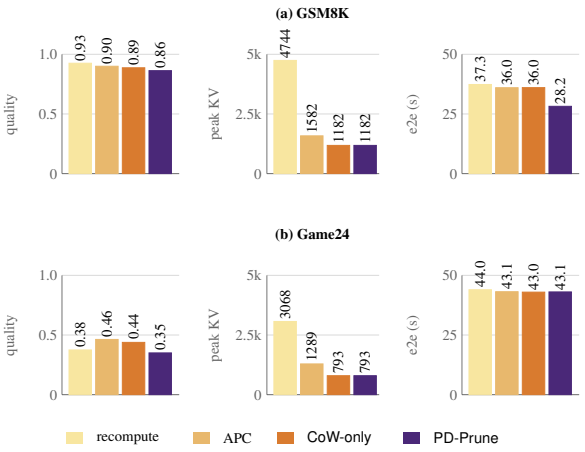


Figure 12: **Four systems on Qwen3-8B** (n=80, tp=2, D=512). Each panel has its own axis; for peak KV and end-to-end time shorter is better. (a) GSM8K. (b) Game of 24. CoW-only and PD-Prune lower peak KV to 0.75× (GSM8K) and 0.62× (Game of 24) APC; the end-to-end reduction of PD-Prune on GSM8K (36.0 to 28.2 s) is due to the answer-marker stop, which is orthogonal and is not attributed to PD-Prune.

on AIME. We do not compare end-to-end time here: the APC baseline decodes the full budget for every item, whereas ours stops at EOS, so time differences would reflect the drivers rather than the methods. The trunk token counts of the two setups also differ slightly (e.g., 95 vs. 73 on R1-8B MATH-500) because of prompt templating, which adds an offset to the ratios.

Table 13: Contest problems, D=2048, k=4, tp=2. All rows of a model come from the same run.

Setting	System	peak	/ APC	acc.
R1-8B MATH	APC	1863	—	0.560
	CoW-only	1599	0.86	0.530
	PD-Prune	1599	0.86	0.530
R1-8B AIME	APC	2118	—	0.041
	CoW-only	1662	0.78	0.041
	PD-Prune	1662	0.78	0.041
Qwen3-4B MATH	APC	1937	—	0.620
	CoW-only	1770	0.91	0.625
	PD-Prune	1770	0.91	0.600
Qwen3-4B AIME	APC	2150	—	0.027
	CoW-only	1794	0.83	0.041
	PD-Prune	1794	0.83	0.041

D Full Ablation Tables

This appendix gives the full ablations behind §5. All runs use Qwen3-8B (tp=1, temperature 0, APC) on the branching mix of §5: GSM8K with n=16, D=512, and MATH-500 (first 32 problems of level ≥ 4) with D=1024. We report computed prefill W (prompt minus cache hits), decode tokens D^+, wall-clock T_{e2e} in seconds (trunk warm-up plus fan-out), committed accuracy (comm.), and any-finished coverage (any).

Notation for this appendix.

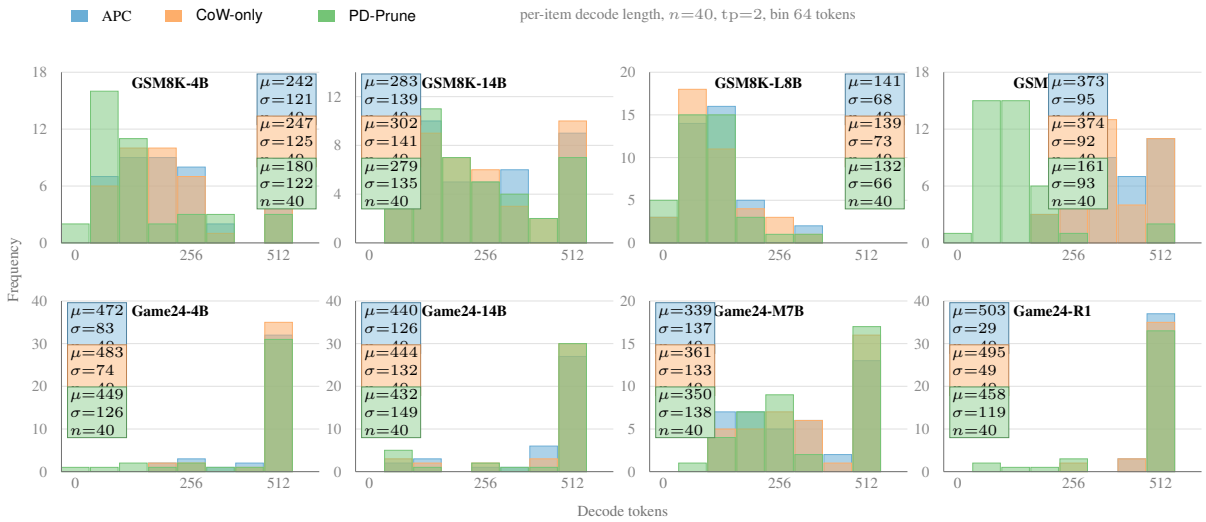


Figure 13: **Per-item decode-length histograms** (n=40, tp=2, D=512, bin width 64). GSM8K (top) and Game of 24 (bottom) overlay APC, CoW-only, and PD-Prune; boxes report μ, σ, and n. Llama-3-8B on Game of 24 is omitted because every output has length 1; Mistral-7B is shown instead.

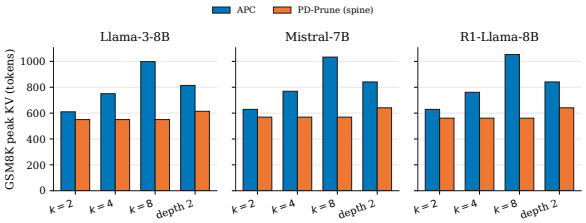


Figure 14: **GSM8K peak KV versus branching** (n=16, tp=2, D=512). The spine peak (PD-Prune and CoW-only) is invariant in k, while the APC peak grows with k.

Symbol	Domain	Meaning
k	$\{4, 8, 16, 32\}$	fan-out width
τ_{pre}	$\{0, 0.15, 0.45\}$	rejection threshold on $\text{Score}_{\text{pre}}(\rho_i)$
τ_{dec}	0.45	extension threshold on $\text{Score}_{\text{dec}}(y_i)$
t_0	$\{64, 128\}$	probe length (MATH-500)
admit	$\mathbb{N}$	residuals that are admitted or probed
W	tokens	computed prefill
D^+	tokens	decode tokens issued
T_{e2e}	s	end-to-end wall-clock
pruner	{ESC, SR, DPTS}	decode-time pruner

D.1 Pruner $\times$ τ_{pre} $\times$ k (GSM8K)

Table 14 is the complete pruner $\times$ threshold $\times$ width matrix. At $k \leq 16$ the three base pruners report identical W, since pruners act after prefill (§4.3); at k=32 measured W diverges slightly across pruners (10528/10176/9856). Rejecting repetitive residuals (τ_{pre}=0.15) removes a quarter of the residuals and keeps coverage within one problem of the pruner-only baseline. Blanket, aggressive prefill-side rejection at 0.45, removes half of the residuals and is faster, but lowers coverage at k=4 and k=8 for ESC.

Analysis. At $k \leq 16$, the three base pruners report identical W and admit counts. Admission changes only that column: $3k/4$ residuals are admitted at τ_{pre}=0.15 (valid and malformed; repetitive residuals rejected) and $k/2$ at 0.45 (valid only). Relative T_{e2e} cuts under ESC+0.15 grow with k: 5%, 6%, 12%, and 17% at k=4, 8, 16, 32. Because SR and DPTS already terminate repetitive residuals during decoding, the same threshold yields smaller relative reductions (6% and 3% at k=16).

D.2 Prefill-only ESC bars

Table 15 isolates the prefill-side score: no decode-time pruning criterion is applied, and every admitted residual is decoded to D. Since $\text{Score}_{\text{pre}}$ takes three values on the mix (repetitive 0.02, malformed 0.22, valid $\approx$ 1), any $\tau_{\text{pre}} \in (0.02, 0.22]$ matches 0.15 and any $\tau_{\text{pre}} \in (0.22, 1)$ matches 0.45. At k=16, the τ_{pre}=0 row coincides in configuration with the ESC baseline of Table 14; committed accuracy across such runs ranges from 8/16 to 11/16.

D.3 Decode-only, prefill-only, and combined (k=16)

Table 16 factorizes the k=16 results. "Stopped" denotes the number of admitted residuals terminated by the pruner's decode-time criterion. Combined rows apply admission first and then the pruner criterion to the admitted set.

Table 14: Full decode-time pruner ablation on GSM8K (Qwen3-8B, branching mix, n=16, D=512). Each cell: any / comm. / T_{e2e} s / W / admit. Base rows have no admission; other rows reject $s_i < \tau_{\text{pre}}$ and admit the rest.

Pruner	Gate	k=4	k=8	k=16	k=32
ESC	base	16/16, 8/16, 8.63, 1320, 4	16/16, 8/16, 10.91, 2624, 8	16/16, 9/16, 16.49, 5008, 16	16/16, 9/16, 32.80, 10528, 32
ESC	τ_{pre}=0.15	15/16, 9/16, 8.23, 1018, 3	16/16, 10/16, 10.26, 2020, 6	16/16, 9/16, 14.45, 3848, 12	16/16, 9/16, 27.15, 7968, 24
ESC	τ_{pre}=0.45	14/16, 9/16, 7.91, 732, 2	15/16, 8/16, 8.67, 1448, 4	16/16, 9/16, 10.95, 2768, 8	16/16, 9/16, 16.60, 5408, 16
SR	base	13/16, 9/16, 8.21, 1320, 4	15/16, 9/16, 9.64, 2624, 8	16/16, 9/16, 13.29, 5008, 16	16/16, 10/16, 26.89, 10176, 32
SR	τ_{pre}=0.15	12/16, 9/16, 8.05, 1018, 3	15/16, 9/16, 9.39, 2020, 6	16/16, 9/16, 12.53, 3848, 12	16/16, 9/16, 24.81, 7744, 24
SR	τ_{pre}=0.45	14/16, 9/16, 7.91, 732, 2	15/16, 8/16, 8.69, 1448, 4	16/16, 9/16, 10.95, 2768, 8	16/16, 9/16, 16.59, 5408, 16
DPTS	base	13/16, 10/16, 8.03, 1320, 4	15/16, 10/16, 9.06, 2624, 8	16/16, 10/16, 11.87, 5008, 16	16/16, 9/16, 23.20, 9856, 32
DPTS	τ_{pre}=0.15	13/16, 7/16, 7.98, 1018, 3	15/16, 10/16, 8.96, 2020, 6	16/16, 10/16, 11.57, 3848, 12	16/16, 10/16, 21.01, 7568, 24
DPTS	τ_{pre}=0.45	14/16, 9/16, 7.92, 732, 2	15/16, 8/16, 8.67, 1448, 4	16/16, 9/16, 10.96, 2768, 8	16/16, 9/16, 16.61, 5408, 16

Table 15: Prefill-only admission (no decode-time cut; every admitted residual decodes to D).

k	τ_{pre}	admit	W	D^+	T_{e2e}	any	comm.
4	0.00	4	1320	32768	8.62	16/16	9/16
4	0.15	3	1018	24576	8.22	16/16	8/16
4	0.45	2	732	16384	7.90	14/16	9/16
8	0.00	8	2624	65536	10.90	16/16	8/16
8	0.15	6	2020	49152	10.25	16/16	10/16
8	0.45	4	1448	32768	8.66	15/16	8/16
16	0.00	16	5008	131072	16.48	16/16	9/16
16	0.15	12	3848	98304	14.45	16/16	10/16
16	0.45	8	2768	65536	10.95	16/16	8/16
32	0.00	32	10528	262144	32.78	16/16	9/16
32	0.15	24	7968	196608	27.14	16/16	9/16
32	0.45	16	5408	131072	16.61	16/16	9/16

D.4 Threshold-Based versus Top-1-Only Admission

Two policies share τ=0.45 but differ in how many residuals are admitted. *Score* admits every i with $s_i \geq \tau$ and index 0 ($k/2$ residuals on the mix). *Top-1 only* admits index 0 alone, the current highest-ranked candidate (W=414). Table 17 shows that Top-1-only admission lowers coverage (ESC 11/16, Speculative Rejection 9/16 at k=16), while score-based admission keeps 16/16.

D.5 MATH-500: Blanket Rejection, Reject-Only Admission, and Probing

Table 18 compares the full fan-out, Blanket, reject-only admission, and truncating repetitive residuals after 128 decode tokens. Table 19 probes the band $[\tau_{\text{pre}}, \tau_{\text{dec}})$ for t_0 tokens and extends to D iff $s'_i \geq \tau_{\text{dec}}$=0.45.

Summary. (i) Blanket rejection of every $s_i < \tau_{\text{dec}}$, i.e., aggressive prefill-side rejection, loses coverage: 20/32 falls to 13/32 at k=16. (ii) Rejecting only repetitive residuals costs at most one problem at $k \leq 8$ but reaches only 16/32 at k=16. (iii) A t_0=128 probe increases coverage to 20/32 at k=16 and cuts 104.10 s to 72.21 s (0.69×); at k=8 the shorter probe t_0=64 is the accuracy-preserving point (16/32 in 42.01 s, exceeding the full fan-out). (iv) Most uncertain residuals pass the decode-side refinement in this mix; probing repetitive residuals as well extends none of them and increases wall-clock time (74.44 s at k=16). (v) At k=16, probe extension recomputes evicted pages, so W exceeds that of the full fan-out although D^+ and wall-clock time decrease; retaining probe pages until the extension decision would eliminate this term.

D.6 Token Accounting Checks

Across Tables 14 and 15, the measured prefill satisfies the predictions of Proposition 1 for a warm trunk: at $k \leq 16$ the three pruners report identical W, and admission removes exactly the computed tokens of the rejected residuals. The control-plane connector table of Table 5 extends to wider fan-outs with the same 25% repetitive-only cut on a stock prefiller and the same alias+gate reduction to the miss of $\mathcal{K}$ (2240/4800/9920 tokens at k=8/16/32). Because residual lengths differ across classes, the reductions are not fixed fractions of W:

	k=4	k=8	k=16	k=32
W, no admission	1320	2624	5008	10528
W, τ_{pre}=0.15	1018 (0.77)	2020 (0.77)	3848 (0.77)	7968 (0.76)
W, τ_{pre}=0.45	732 (0.55)	1448 (0.55)	2768 (0.55)	5408 (0.51)

Repetitive residuals therefore account for about 23% of residual prefill and malformed residuals for a further 22%, although each class is a quarter of the residuals. Combined rows never exceed the prefill-only W at the same τ_{pre}.

E Extended Related Work

Inference-time search. Chain-of-thought prompting elicits intermediate steps [32]; Tree-of-Thoughts and Monte Carlo tree search expand, score, and backtrack over such steps [35, 39]; self-consistency samples complete traces and takes a majority [31]. Accuracy and token cost both

Table 16: Factorization at k=16 (GSM8K, Qwen3-8B). Decode-only: τ_{pre} absent, W=5008. Prefill-only: decode-time cut disabled, decode to D. Combined: both. Any-finished is 16/16 on every row. Prefill-only and combined rows come from separate runs.

Factor	Setting	admit	stopped	W	D^+	T_{e2e}	comm.
Decode-only							
	ESC (D=512)	16	0	5008	131072	16.49	9/16
	SR (τ=256)	16	8	5008	98304	13.29	9/16
	DPTS (d=100)	16	8	5008	78336	11.87	10/16
Prefill-only							
	τ_{pre}=0.15	12	0	3848	98304	14.45	10/16
	τ_{pre}=0.45	8	0	2768	65536	10.95	8/16
Combined							
	ESC+0.15	12	0	3848	98304	14.45	9/16
	SR+0.15	12	4	3848	81920	12.53	9/16
	DPTS+0.15	12	4	3848	71936	11.57	10/16
	ESC+0.45	8	0	2768	65536	10.95	9/16
	SR+0.45	8	0	2768	65536	10.95	9/16
	DPTS+0.45	8	0	2768	65536	10.96	9/16

Table 17: Score-based versus Top-1-only admission at τ=0.45 (GSM8K, Qwen3-8B). Rows come from a separate run from Table 14.

k	Policy	Pruner	admit	W	T_{e2e}	any	comm.
16	base	ESC	16	5008	17.02	16/16	10/16
16	score	ESC	8	2768	10.91	16/16	10/16
16	Top-1 only	ESC	1	414	7.30	11/16	11/16
16	base	SR	16	5008	13.12	16/16	12/16
16	score	SR	8	2768	10.84	16/16	10/16
16	Top-1 only	SR	1	414	7.29	9/16	9/16
16	base	DPTS	16	5008	11.85	16/16	10/16
16	score	DPTS	8	2768	10.89	16/16	9/16
16	Top-1 only	DPTS	1	414	7.30	11/16	11/16
8	Top-1 only	ESC	1	414	7.28	11/16	11/16
8	score	ESC	4	1448	8.65	14/16	9/16
4	Top-1 only	ESC	1	414	7.29	10/16	10/16
4	score	ESC	2	732	7.88	14/16	11/16

grow with the number of traces. EquivPruner observes that textually distinct steps can be the same mathematical action and retains one representative per equivalence class [21]. PD-Prune neither changes the search procedure nor decides semantic equivalence; it decides, for the residuals a branching controller proposes, which are prefilled, transferred, and decoded.

Path pruning. ESC stops sampling once a window agrees [17]; Speculative Rejection halts low-reward partial generations at a horizon [26]; DPTS early-stops children after a mini-step [10]; STOP categorizes pruning signals as internal or external and learned or not, and trains an internal futility token [4]. These methods read generated tokens or compare written steps, so they act on the decode side of a PD deployment. PD-Prune adds a pre-kernel decision and a decode-side probe around them and leaves their decode-time cut unchanged (§C.4); a learned signal such as STOP's could replace $\text{Score}_{\text{dec}}$.

Request-level serving. Orca introduced iteration-level scheduling [37]; vLLM pages the KV cache and hashes full blocks for APC [14]; SGLang shares prefixes through a radix tree [41]; SARATHI and Sarathi-Serve chunk prefills to co-schedule them with decode [3, 2]; DistServe and Splitwise disaggregate prefill and decode [42, 25]. Whether these engines share a trunk across concurrent siblings depends on the cache regime (§3.2); in all cases they serve every request they receive. PD-Prune uses their residency information as an input to admission and their connector as the channel whose volume admission reduces.

Program-, workflow-, and conversation-level scheduling. Autellix schedules by program-level attained service [23]; JITServe allocates just enough bandwidth under heterogeneous SLOs [40]; Hermes, ThunderAgent, and SAGA schedule at program or workflow granularity [22, 13, 11]; Parrot, Ayo, and Pythia exploit application structure [20, 27, 38]. ConServe disaggregates prefill and decode scheduling at conversation granularity based on observed agent behavior [9]. These

Table 18: MATH-500 level ≥ 4, Qwen3-8B, n=32, D=1024. "Blanket" rejects every s_i<0.45. "Reject-only" rejects repetitive residuals only. "short probe@128" admits every residual but limits decoding of repetitive residuals to 128 tokens.

k	Setting	admit	W	D^+	T_{e2e}	any
4	full	4	2844	131072	25.34	12/32
4	blanket	2	1566	65536	19.11	6/32
4	reject-only	3	2189	98304	23.05	11/32
4	short probe@128	4	2844	102400	23.14	10/32
8	full	8	5592	262144	52.50	15/32
8	blanket	4	3036	131072	25.60	11/32
8	reject-only	6	4282	196608	41.53	17/32
8	short probe@128	8	5592	204800	42.26	15/32
16	full	16	12320	524288	104.10	20/32
16	blanket	8	6200	262144	52.05	13/32
16	reject-only	12	9332	393216	71.39	16/32
16	short probe@128	16	11856	409600	74.03	17/32

Table 19: Probe admission on MATH-500. Probe@t_0: repetitive residuals rejected; malformed residuals probed and extended iff $s'_i \geq 0.45$. Probe-rep.: repetitive residuals are probed as well (none is extended). At k=16, W includes the recomputation Δ_{ext} of evicted probe pages; it is about 19.4k tokens at t_0=128 and 11.2k at t_0=64 relative to reject-only (9332).

k	Setting	admit	W	D^+	T_{e2e}	any
4	probe@64	3	2460	98304	23.96	8/32
4	probe@128	3	2460	98304	24.70	9/32
4	probe-rep.@128	4	3115	102400	24.89	10/32
8	probe@64	6	4824	196608	42.01	16/32
8	probe@128	6	4824	196608	43.30	13/32
8	probe-rep.@128	8	6134	204800	43.32	15/32
16	probe@64	12	20544	393216	72.50	18/32
16	probe@128	12	28736	393216	72.21	20/32
16	probe-rep.@128	16	31260	409600	74.44	20/32

systems decide when and where requests run; PD-Prune decides which sibling continuations become requests and at which stage a discarded one stops consuming resources.

KV retention, overlap, and copy-on-write. InferCept, Continuum, and MORI preserve or place context across tool calls [1, 16, 33]. Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]; SmoothAgent transforms long-horizon context; IdleSpec uses idle time for speculative planning to improve accuracy rather than KV reuse [24, 7]. ForkKV shares a base KV cache copy-on-write across multi-LoRA agents [29], which is closest to our context tree; PD-Prune applies copy-on-write to sibling continuations of one agent step and couples it with admission and spine retention. CachePrune edits KV neurons so that a model ignores instructions embedded in retrieved context [28]; it changes attention state within one request and does not decide which siblings are served.

Speculative decoding. Speculative decoding and multi-head drafters propose *output* tokens and verify them against the target model [15, 6, 18]. The object PD-Prune may prefill speculatively is a future *prompt* prefix, such as a known tool-call suffix (Appendix B.3); speculatively prefilled pages are never sampled from, and (18) binds the prompt that actually arrives.